

# BFS Traversal

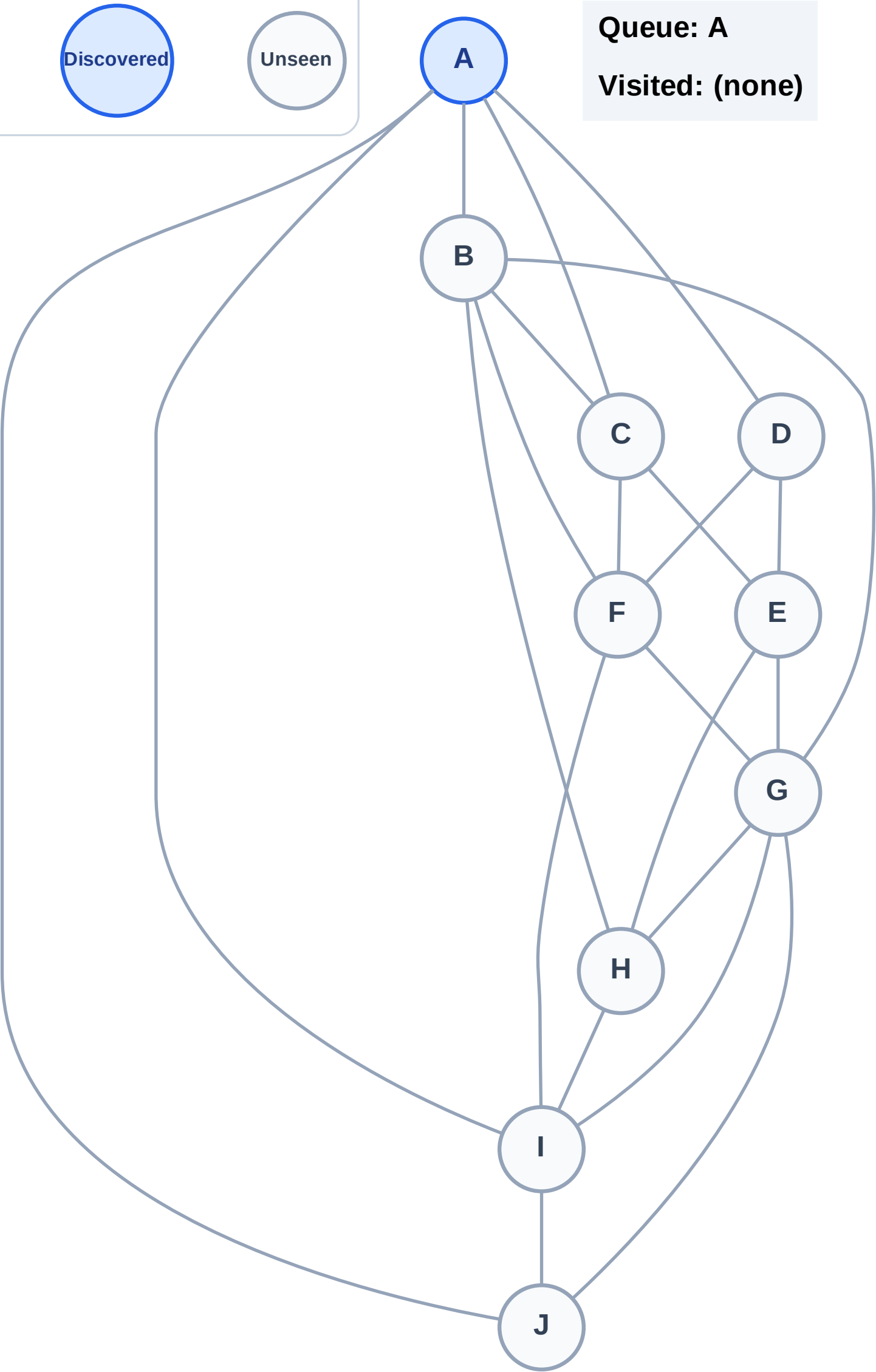
Step 1: Start at A

## Legend



Queue: A

Visited: (none)



# BFS Traversal

Step 2: Process node A

Legend

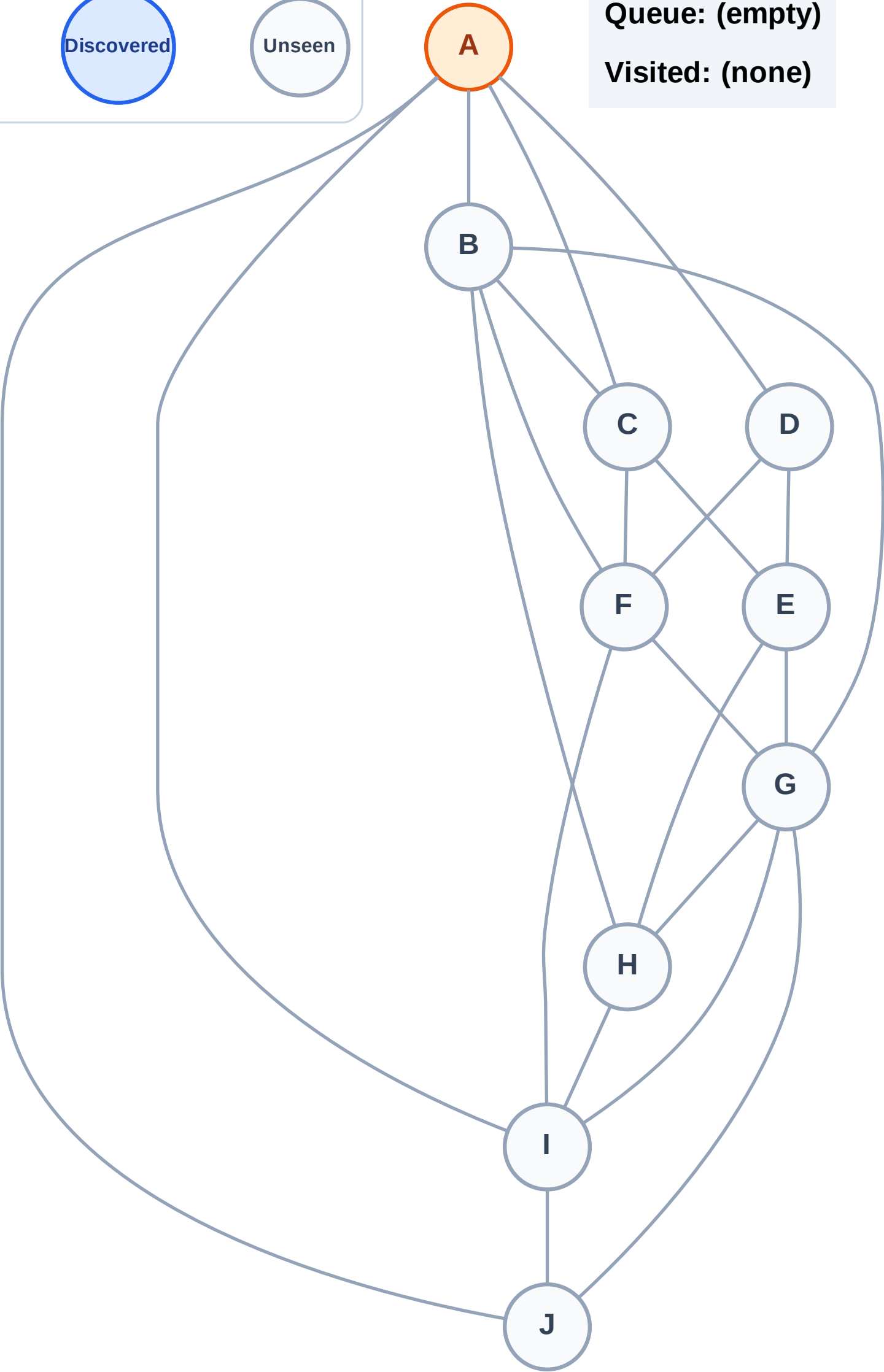
Complete

Processing

Discovered

Unseen

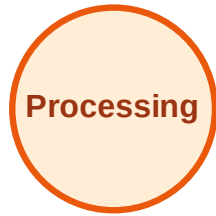
Queue: (empty)  
Visited: (none)



# BFS Traversal

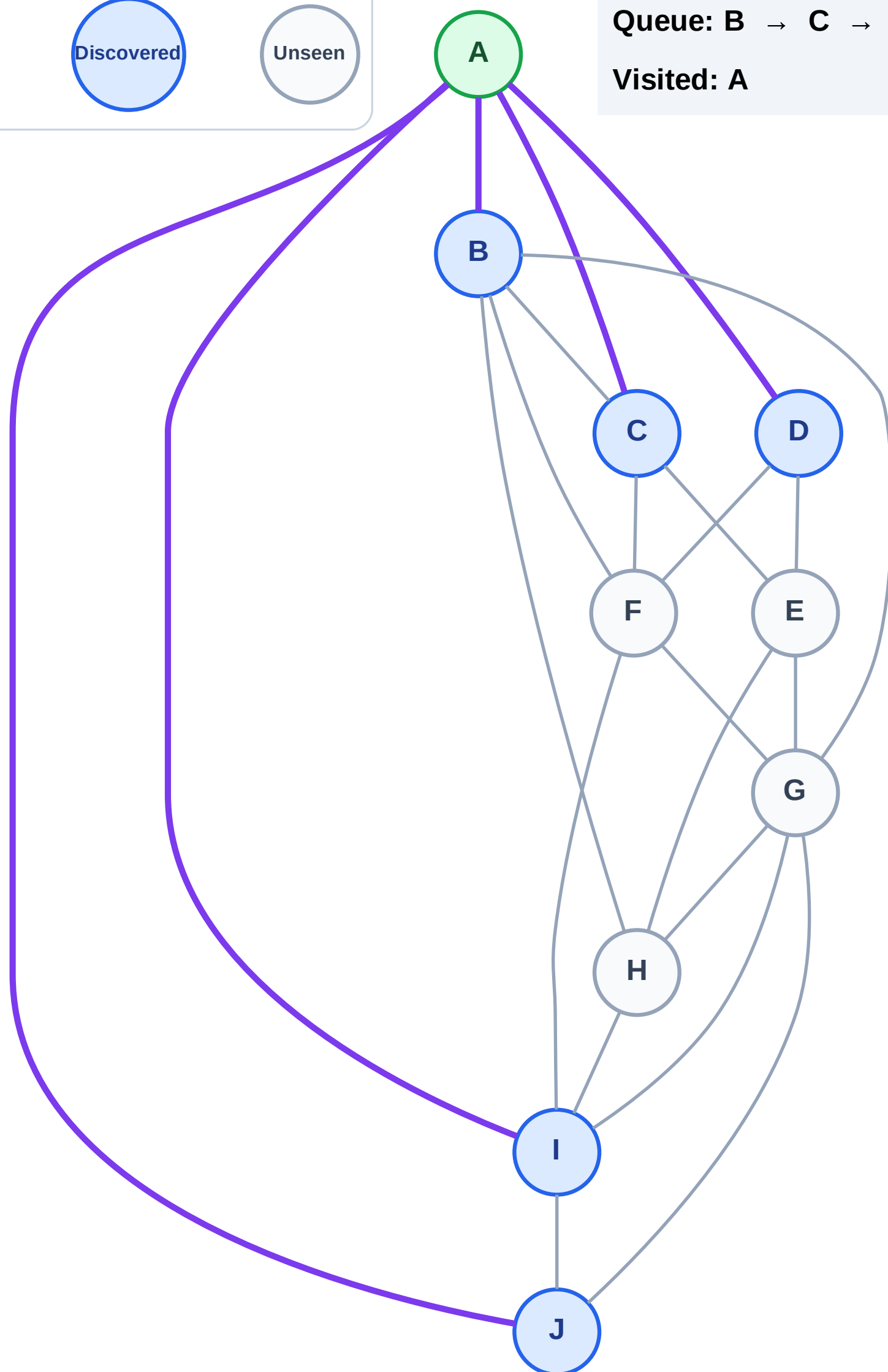
Step 3: Discover B, C, D, I, J from A

## Legend



Queue: B → C → D → I → J

Visited: A



# BFS Traversal

Step 4: Process node B

Legend

Complete

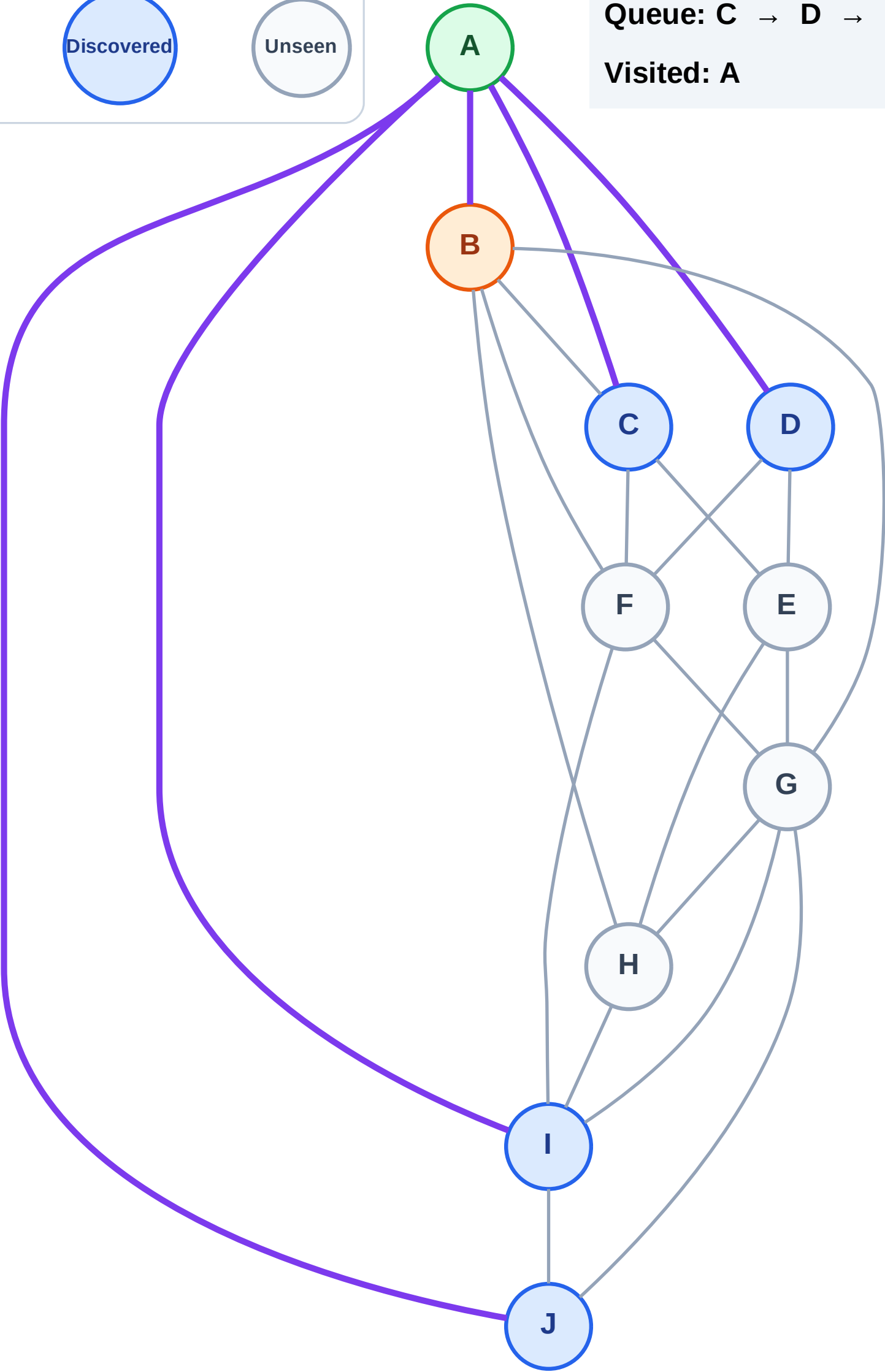
Processing

Discovered

Unseen

Queue: C → D → I → J

Visited: A



# BFS Traversal

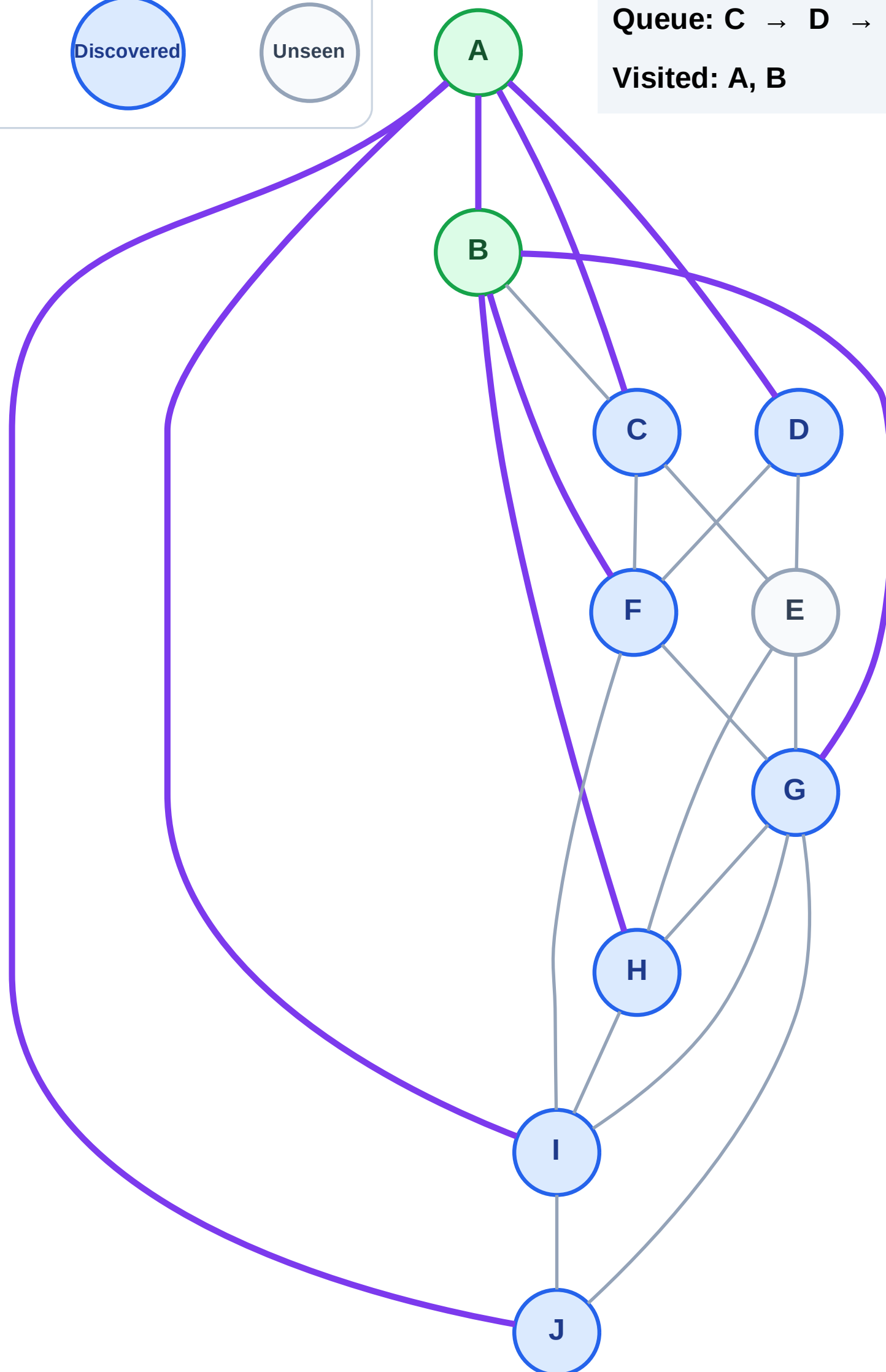
Step 5: Discover F, G, H from B

## Legend



Queue: C → D → I → J → F → G → H

Visited: A, B



# BFS Traversal

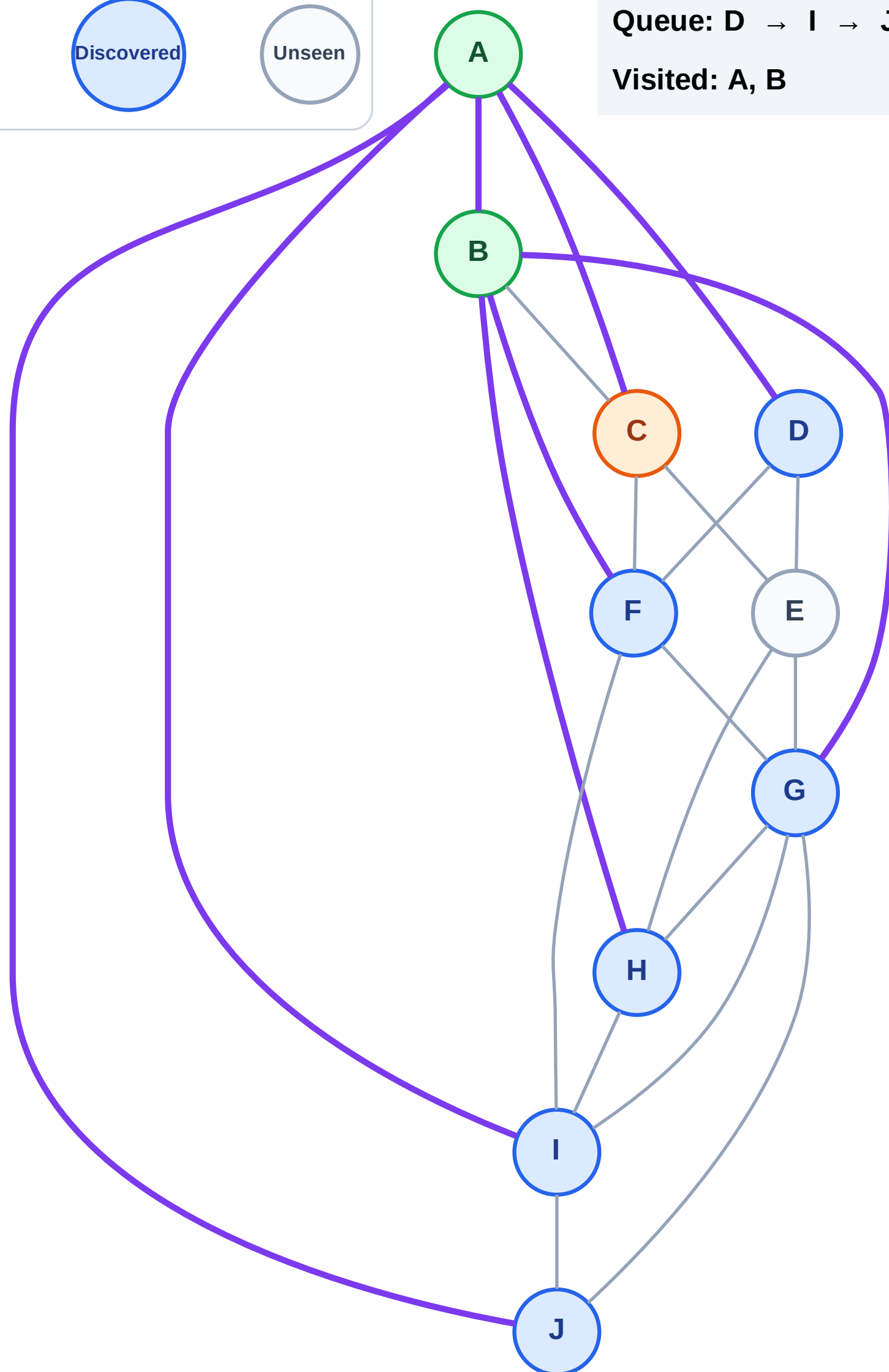
Step 6: Process node C

## Legend



Queue: D → I → J → F → G → H

Visited: A, B



# BFS Traversal

Step 7: Discover E from C

## Legend

Complete

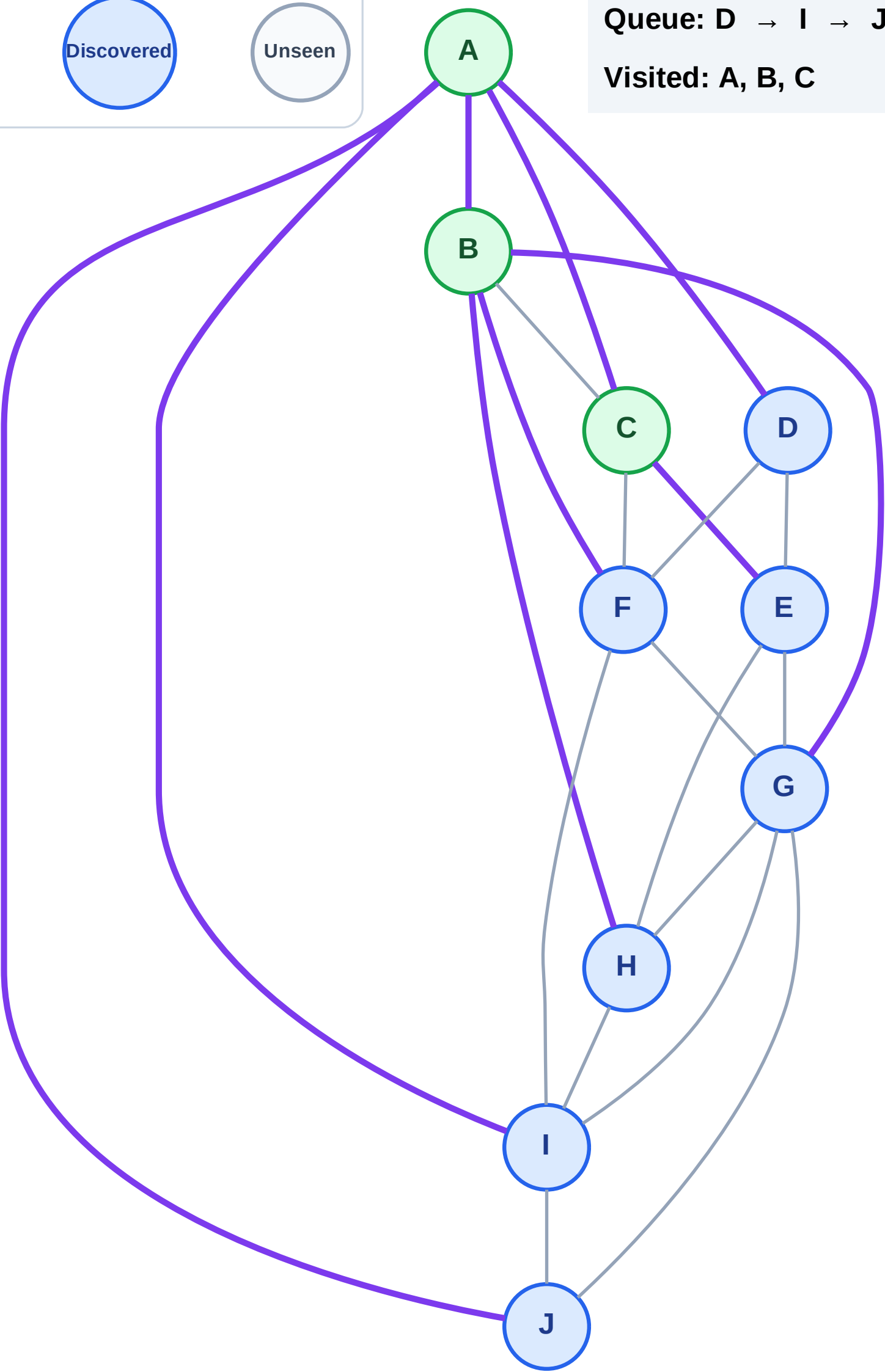
Processing

Discovered

Unseen

Queue: D → I → J → F → G → H → E

Visited: A, B, C



# BFS Traversal

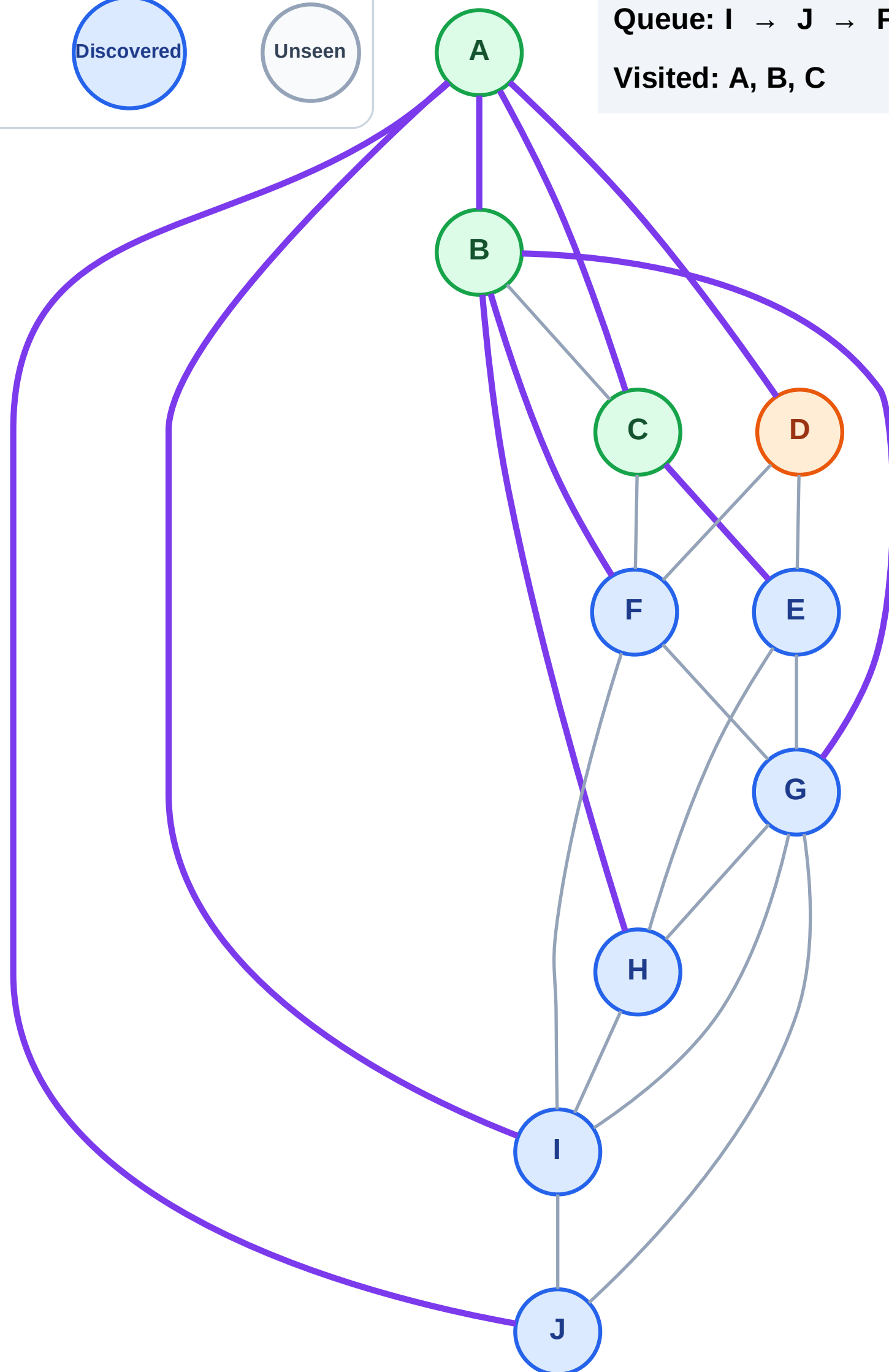
Step 8: Process node D

## Legend



Queue: I → J → F → G → H → E

Visited: A, B, C



# BFS Traversal

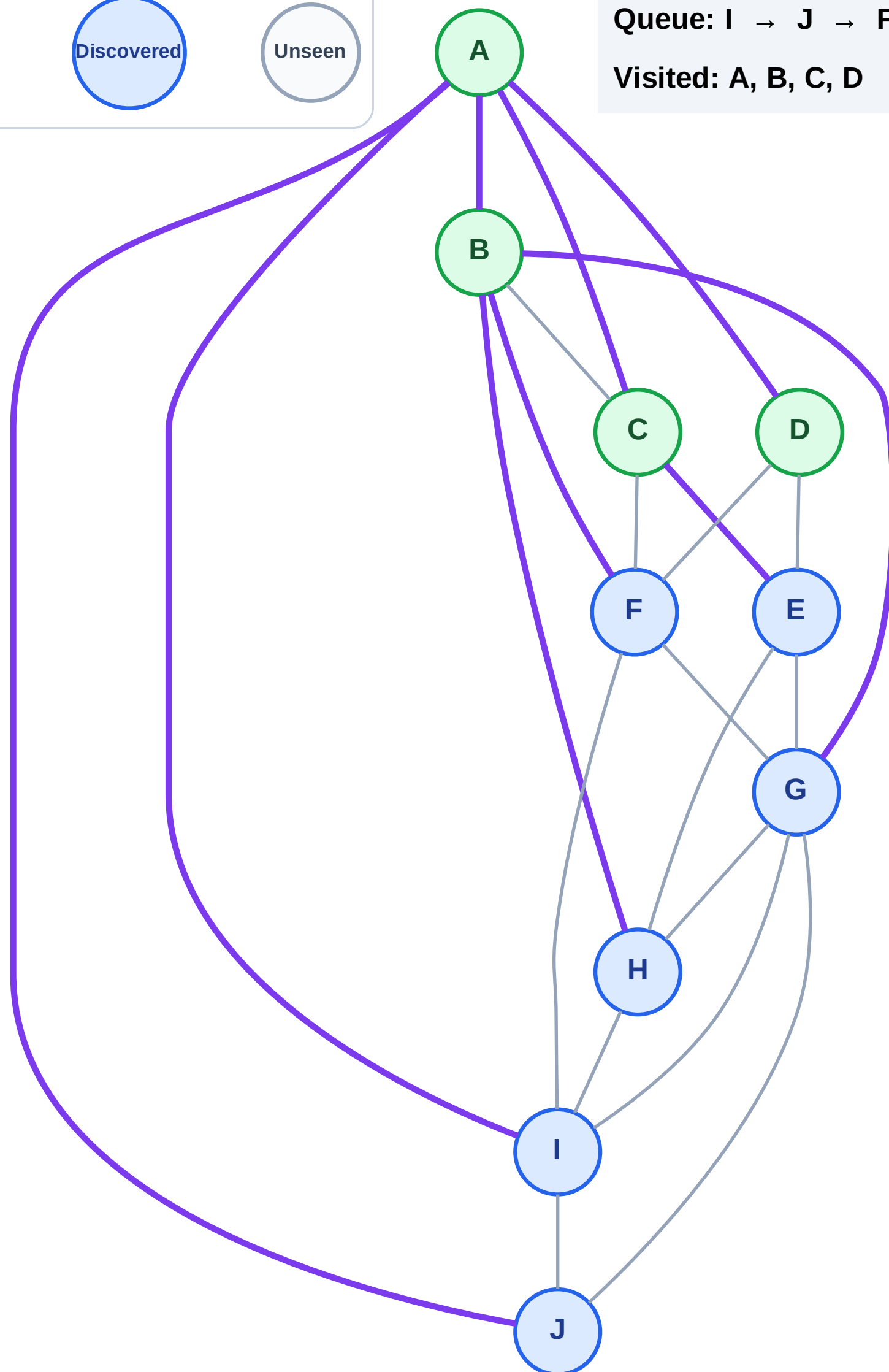
Step 9: No new neighbors from D

## Legend



Queue: I → J → F → G → H → E

Visited: A, B, C, D



# BFS Traversal

Step 10: Process node I

Legend

Complete

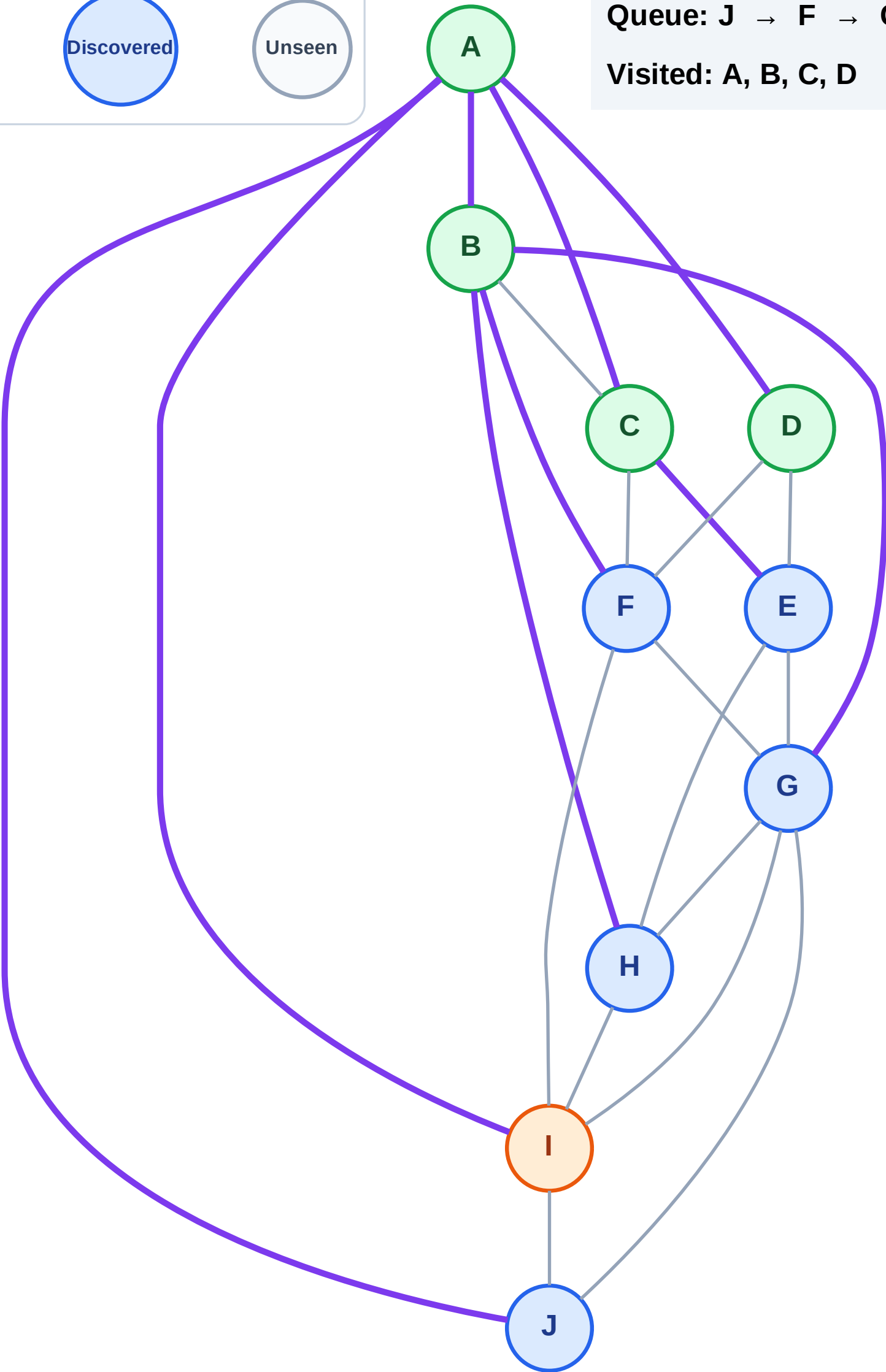
Processing

Discovered

Unseen

Queue: J → F → G → H → E

Visited: A, B, C, D



# BFS Traversal

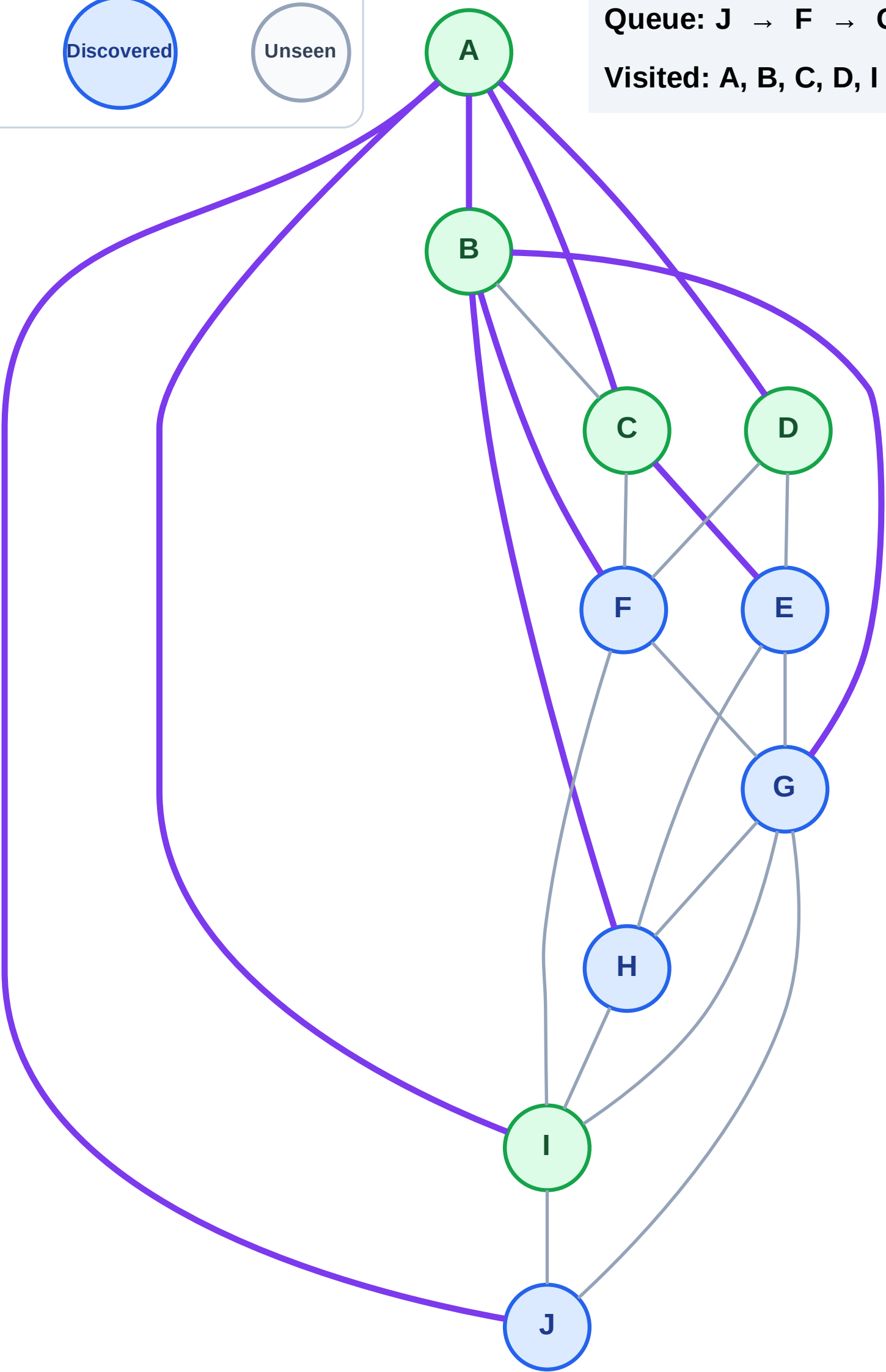
Step 11: No new neighbors from I

## Legend



Queue: J → F → G → H → E

Visited: A, B, C, D, I



# BFS Traversal

Step 12: Process node J

Legend

Complete

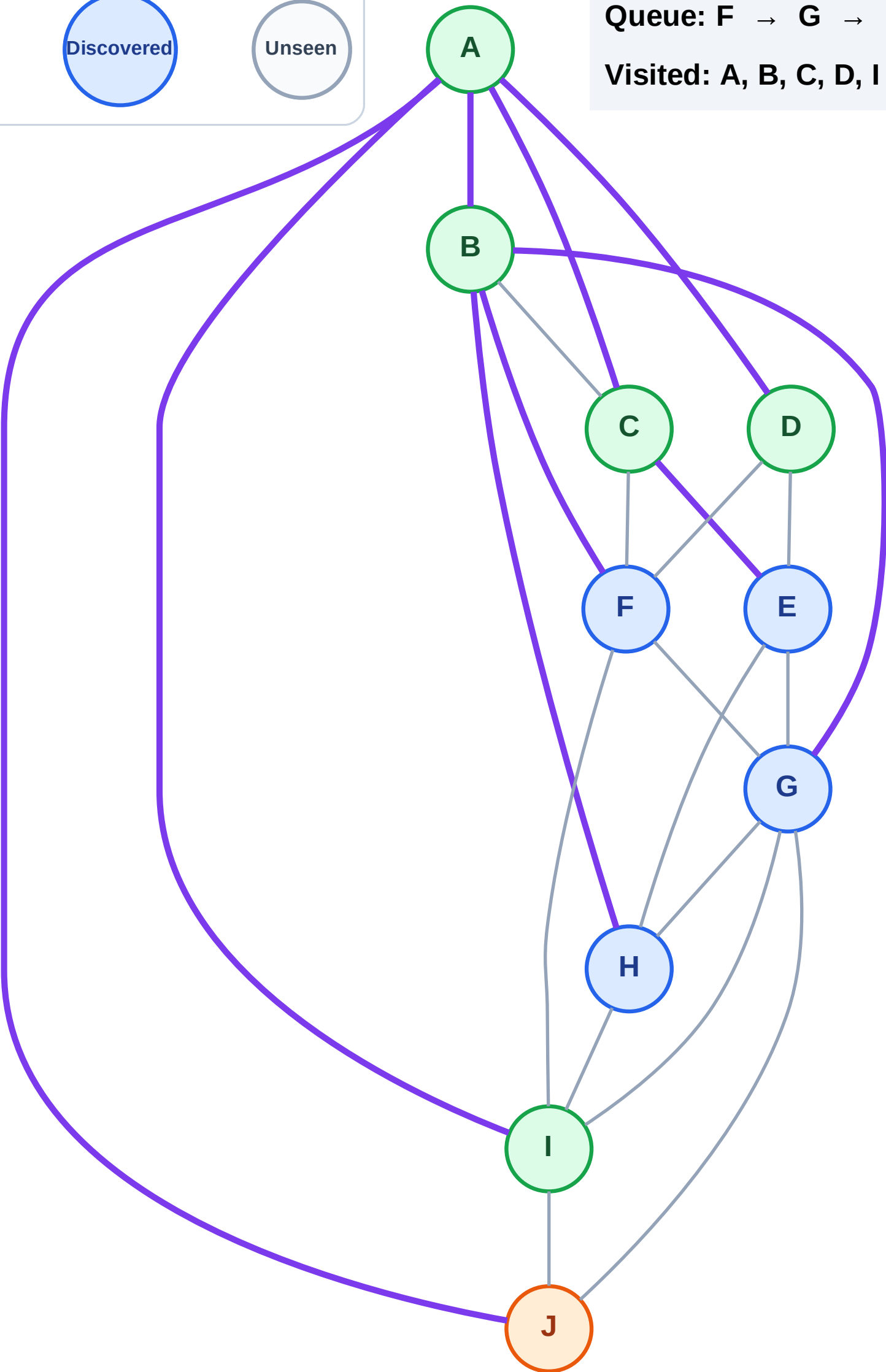
Processing

Discovered

Unseen

Queue: F → G → H → E

Visited: A, B, C, D, I



# BFS Traversal

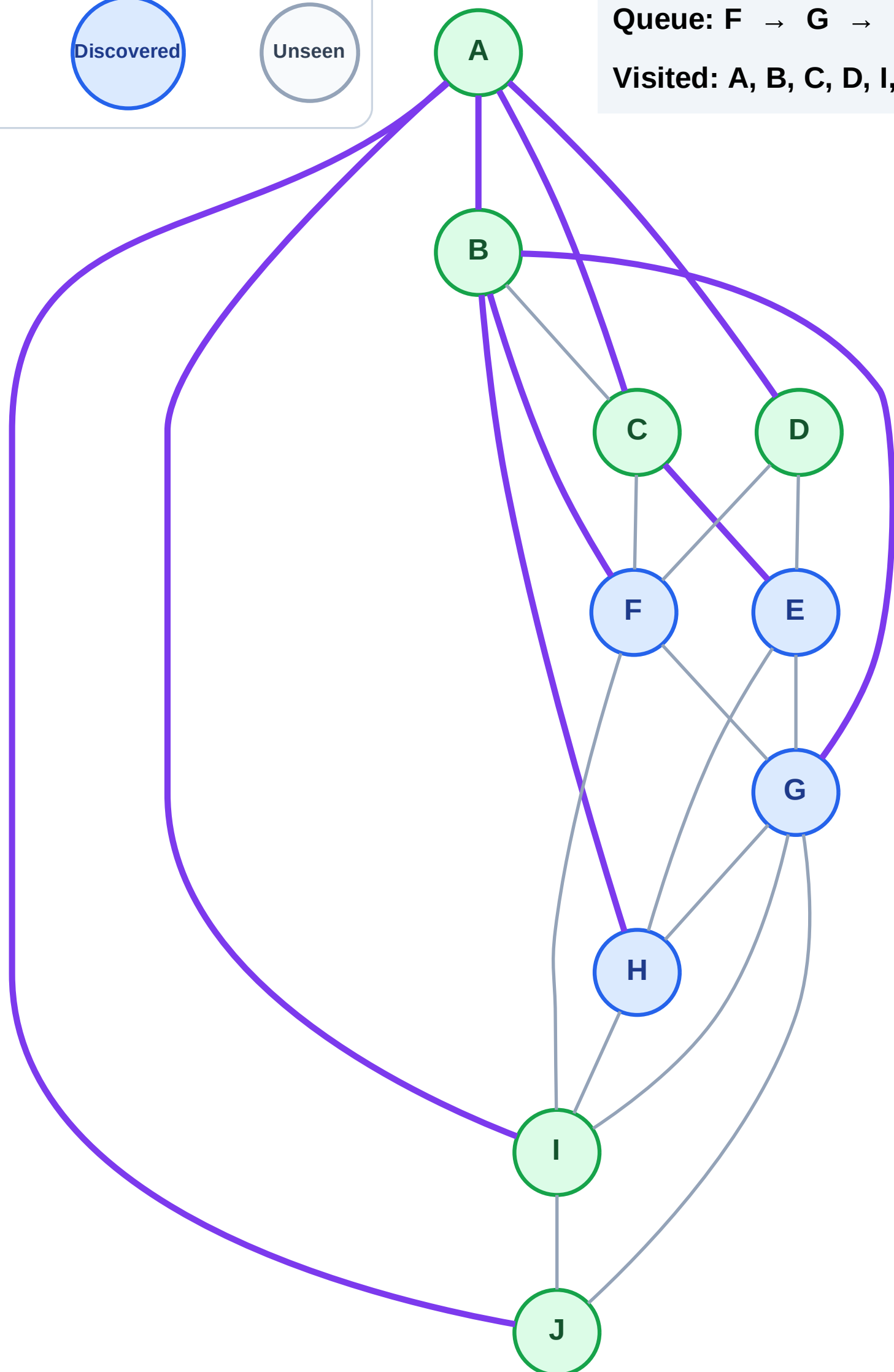
Step 13: No new neighbors from J

## Legend



Queue: F → G → H → E

Visited: A, B, C, D, I, J



# BFS Traversal

Step 14: Process node F

## Legend



Complete



Processing



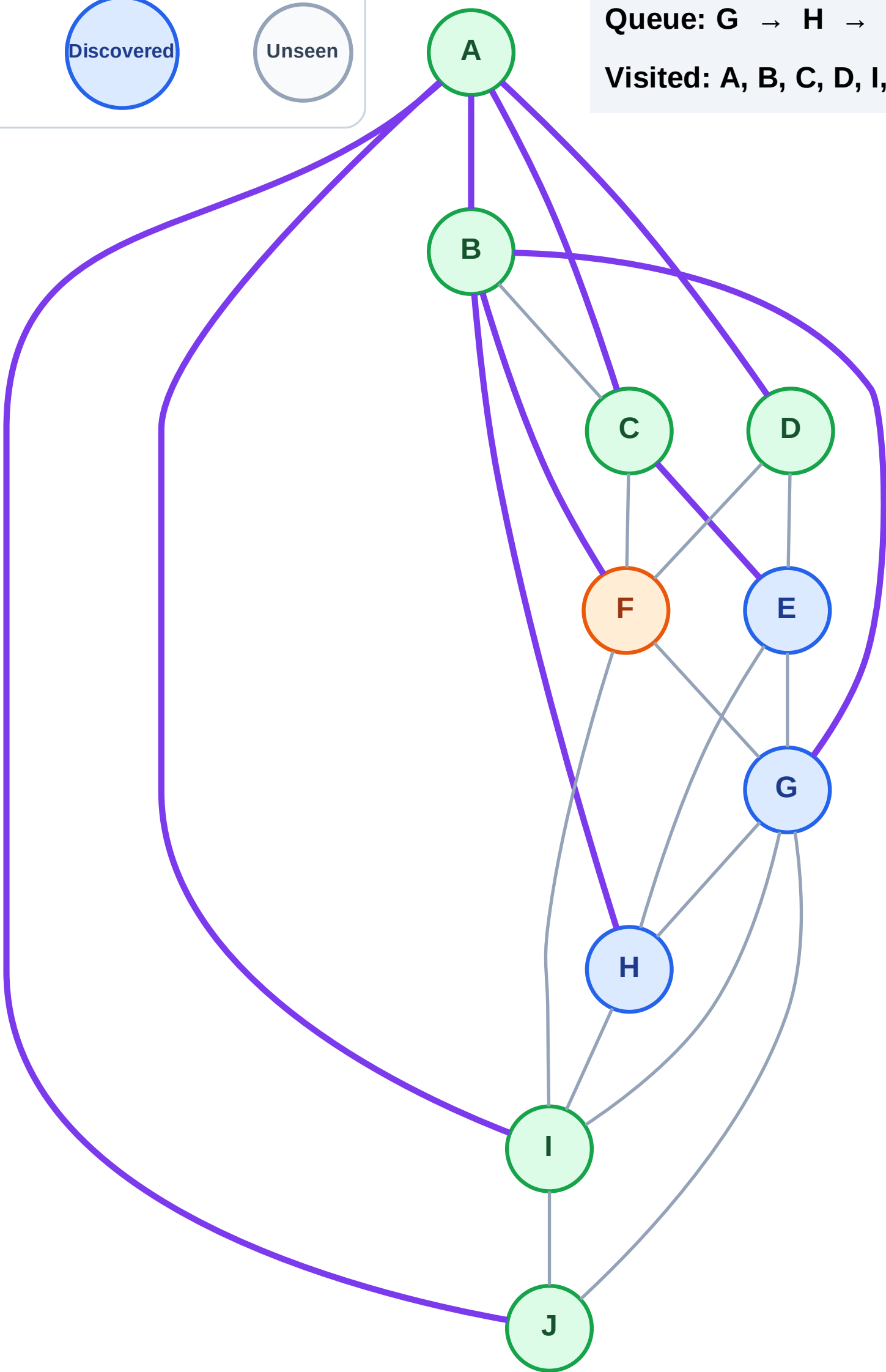
Discovered



Unseen

Queue: G → H → E

Visited: A, B, C, D, I, J



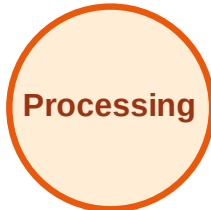
# BFS Traversal

Step 15: No new neighbors from F

## Legend



Complete



Processing



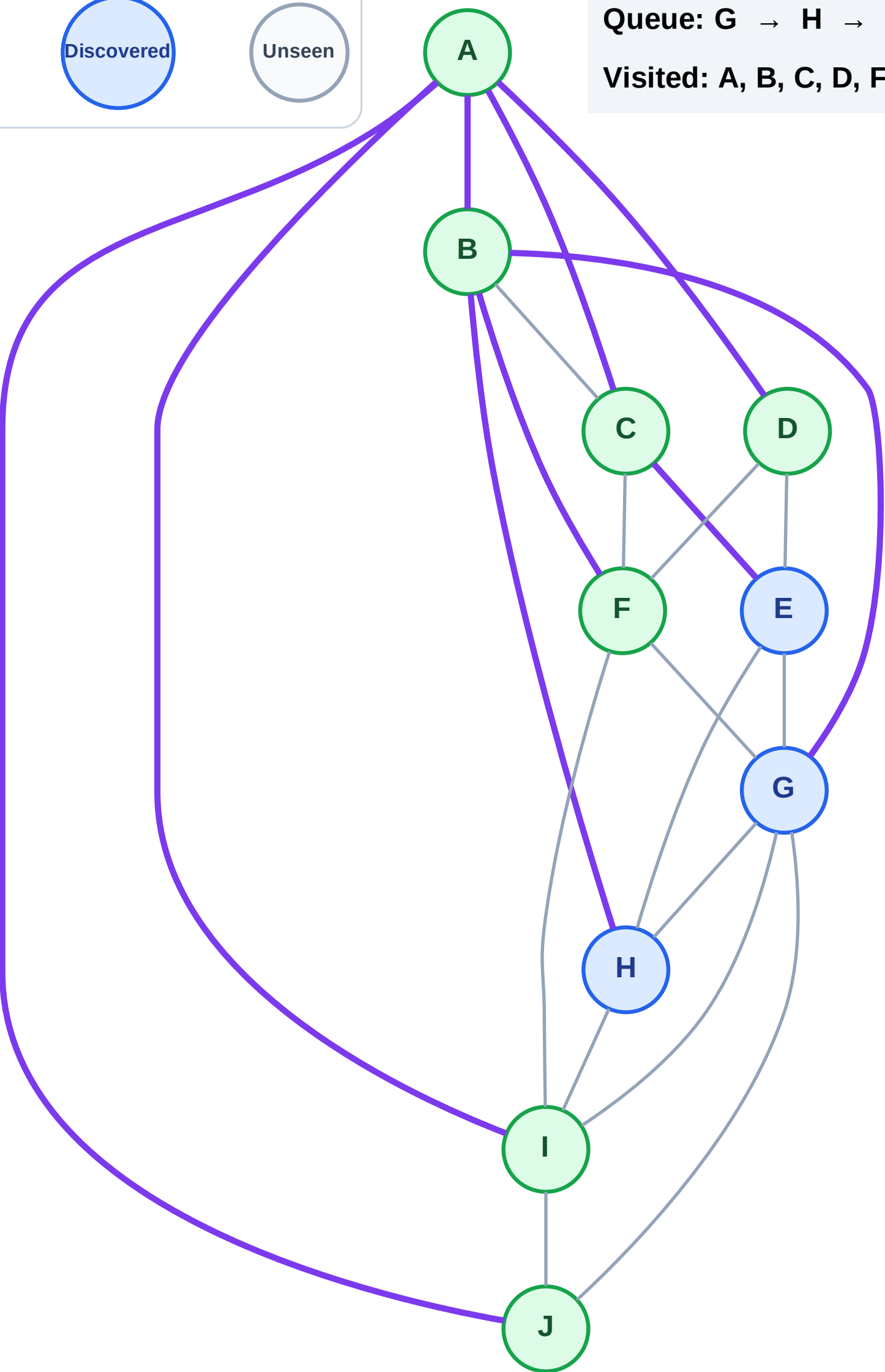
Discovered



Unseen

Queue: G → H → E

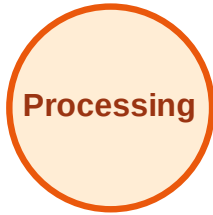
Visited: A, B, C, D, F, I, J



# BFS Traversal

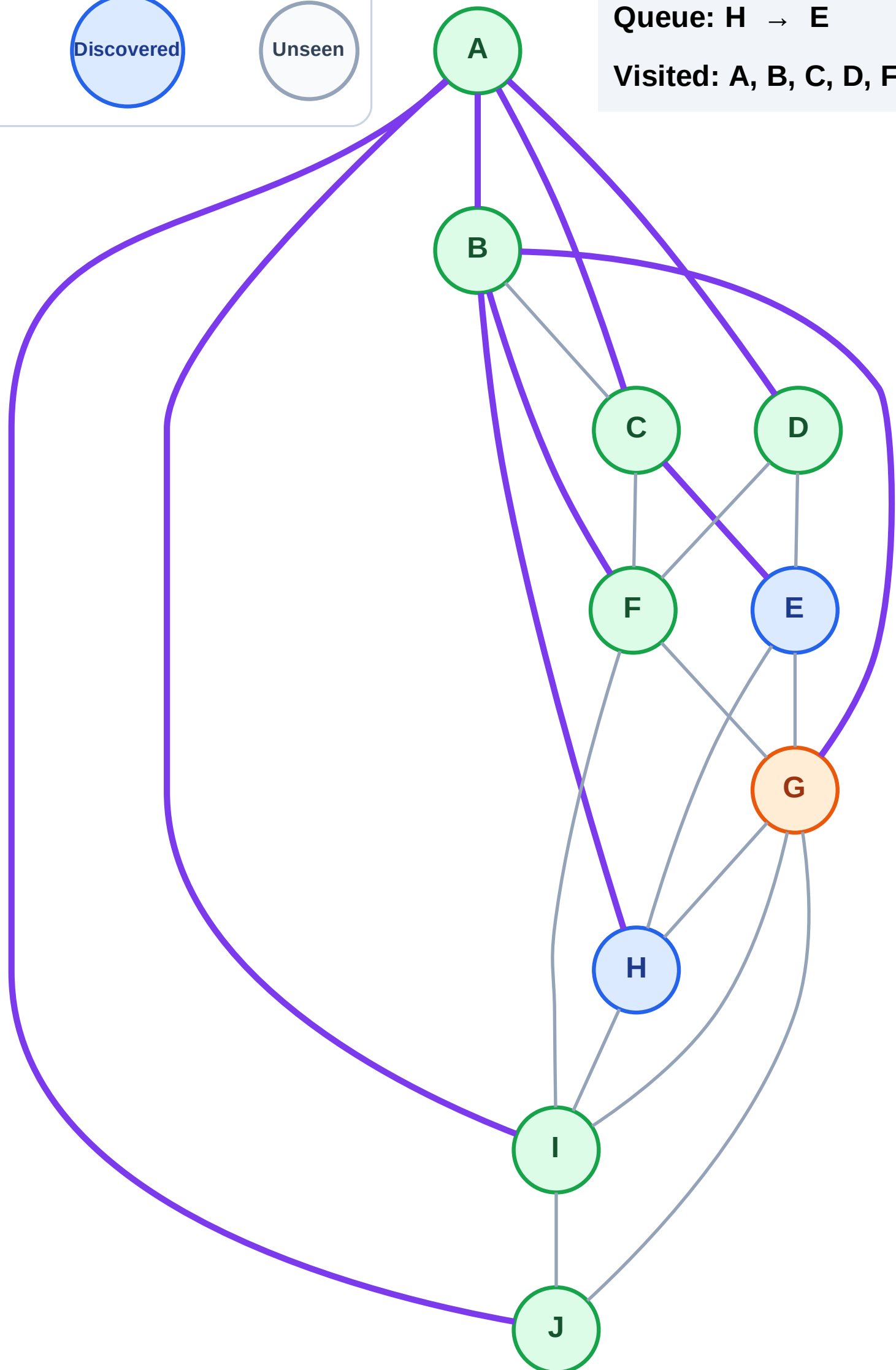
Step 16: Process node G

## Legend



Queue: H → E

Visited: A, B, C, D, F, I, J



# BFS Traversal

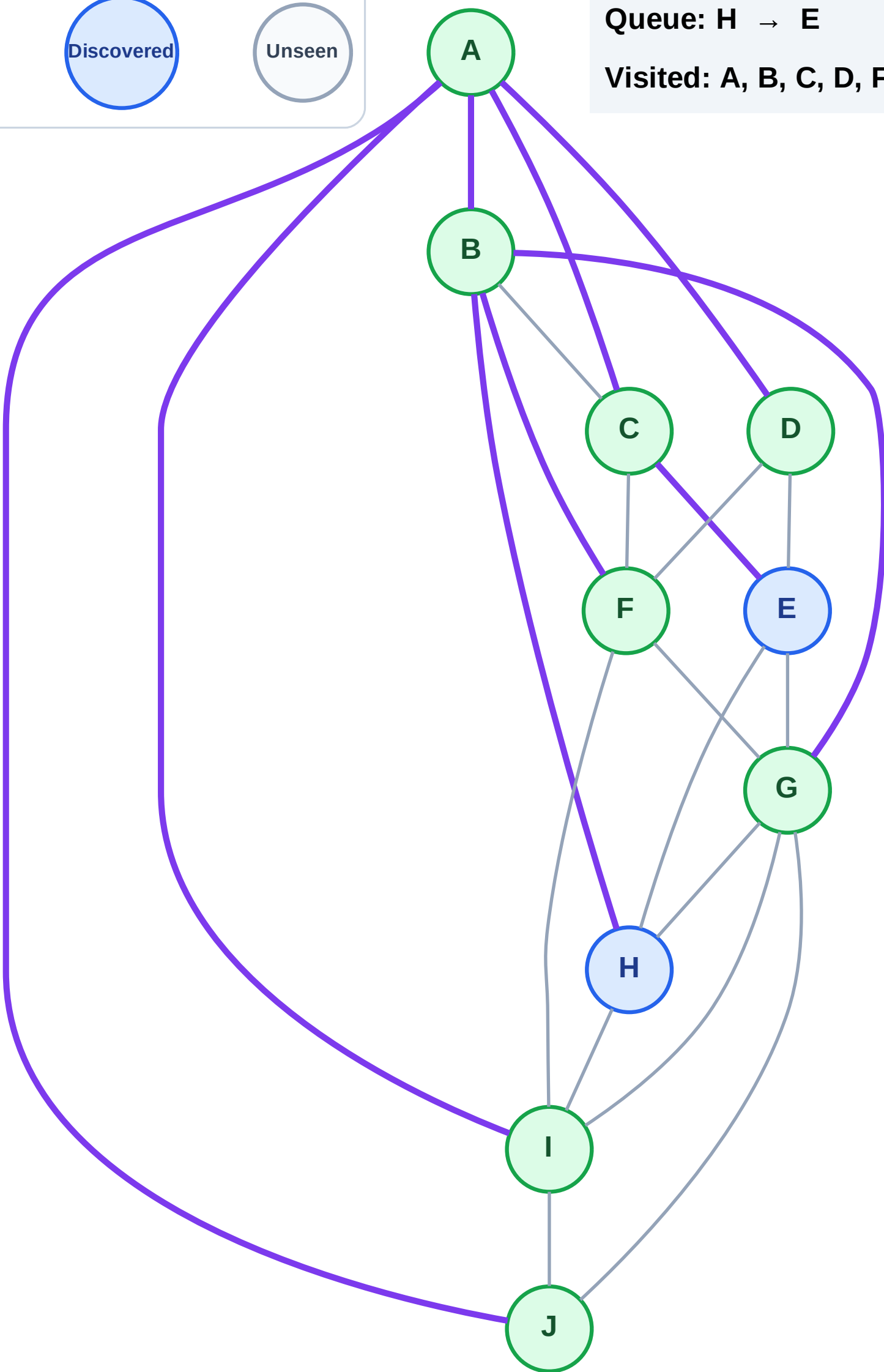
Step 17: No new neighbors from G

## Legend



Queue: H → E

Visited: A, B, C, D, F, G, I, J



# BFS Traversal

Step 18: Process node H

Legend

Complete

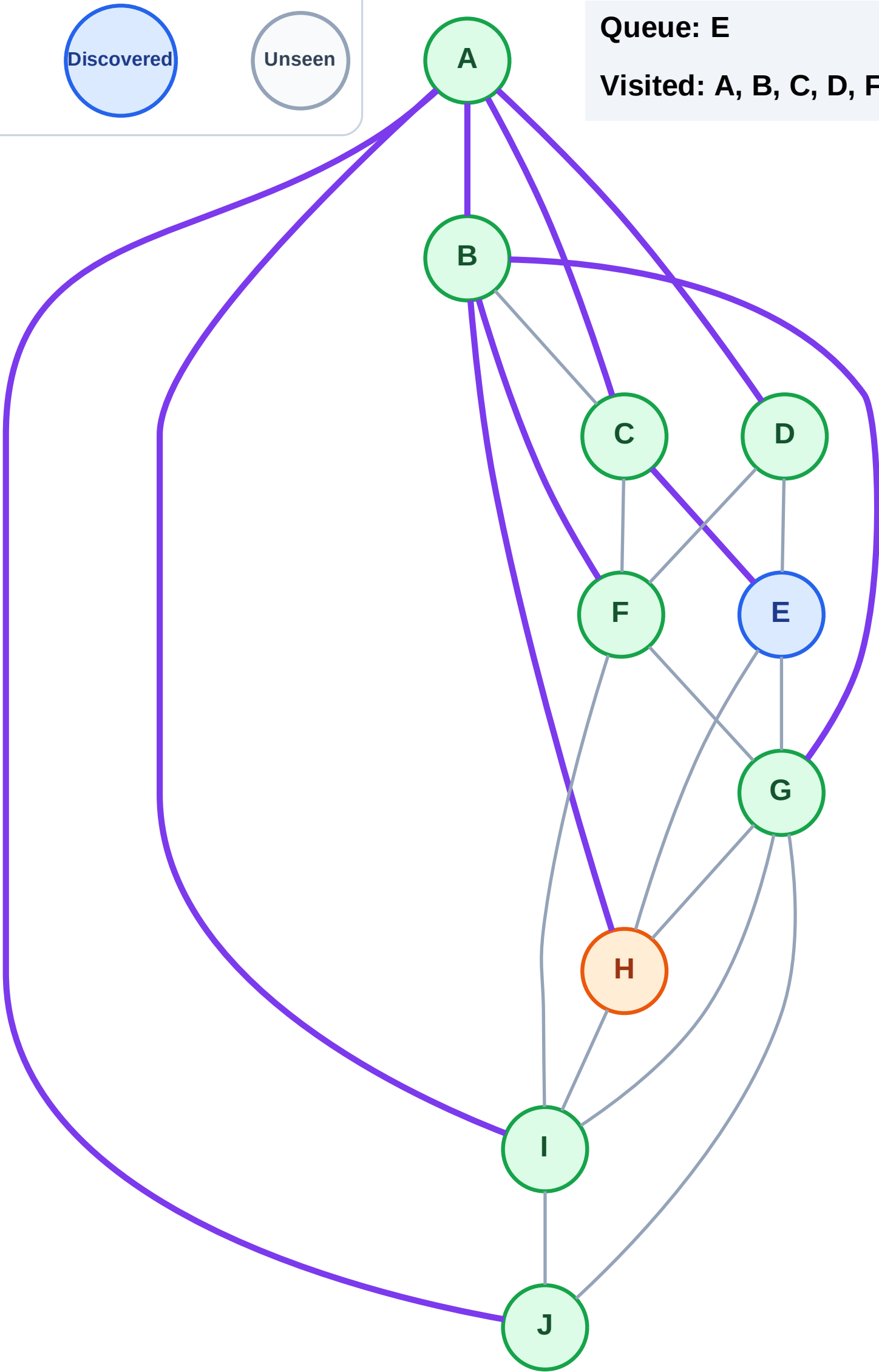
Processing

Discovered

Unseen

Queue: E

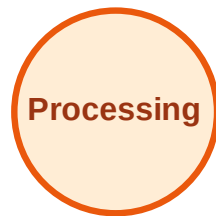
Visited: A, B, C, D, F, G, I, J



# BFS Traversal

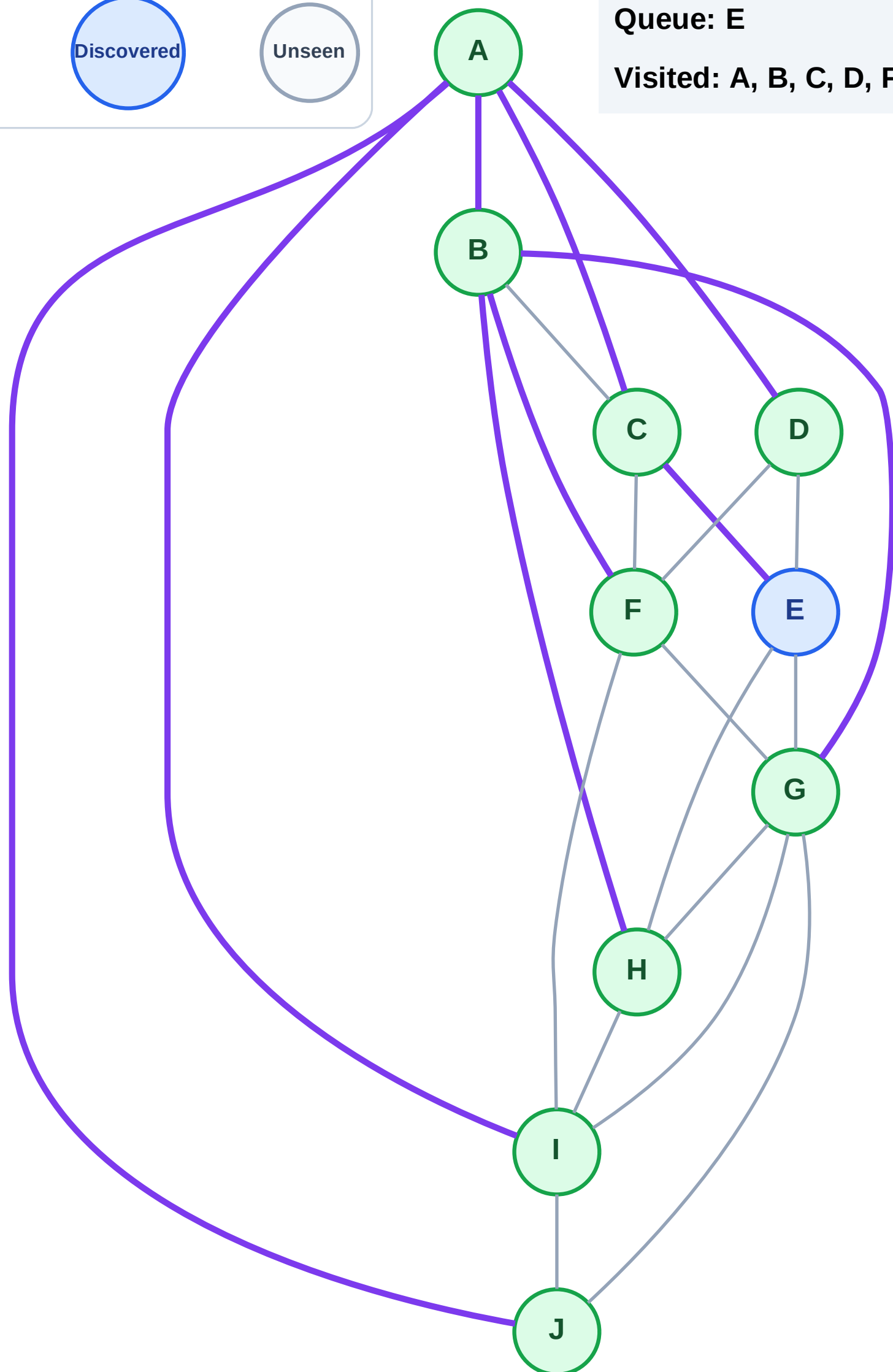
Step 19: No new neighbors from H

## Legend



Queue: E

Visited: A, B, C, D, F, G, H, I, J



# BFS Traversal

Step 20: Process node E

Legend

Complete

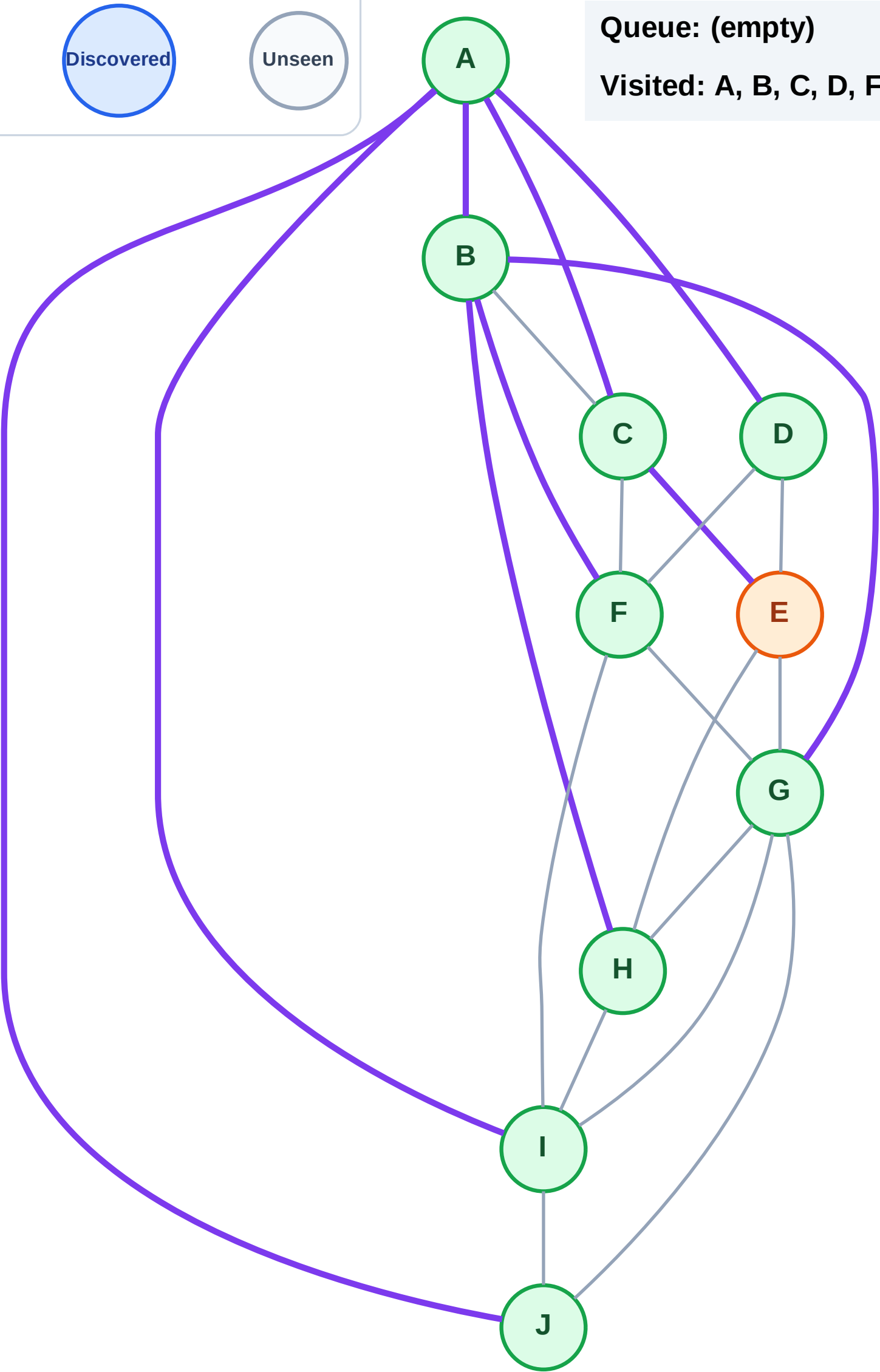
Processing

Discovered

Unseen

Queue: (empty)

Visited: A, B, C, D, F, G, H, I, J



# BFS Traversal

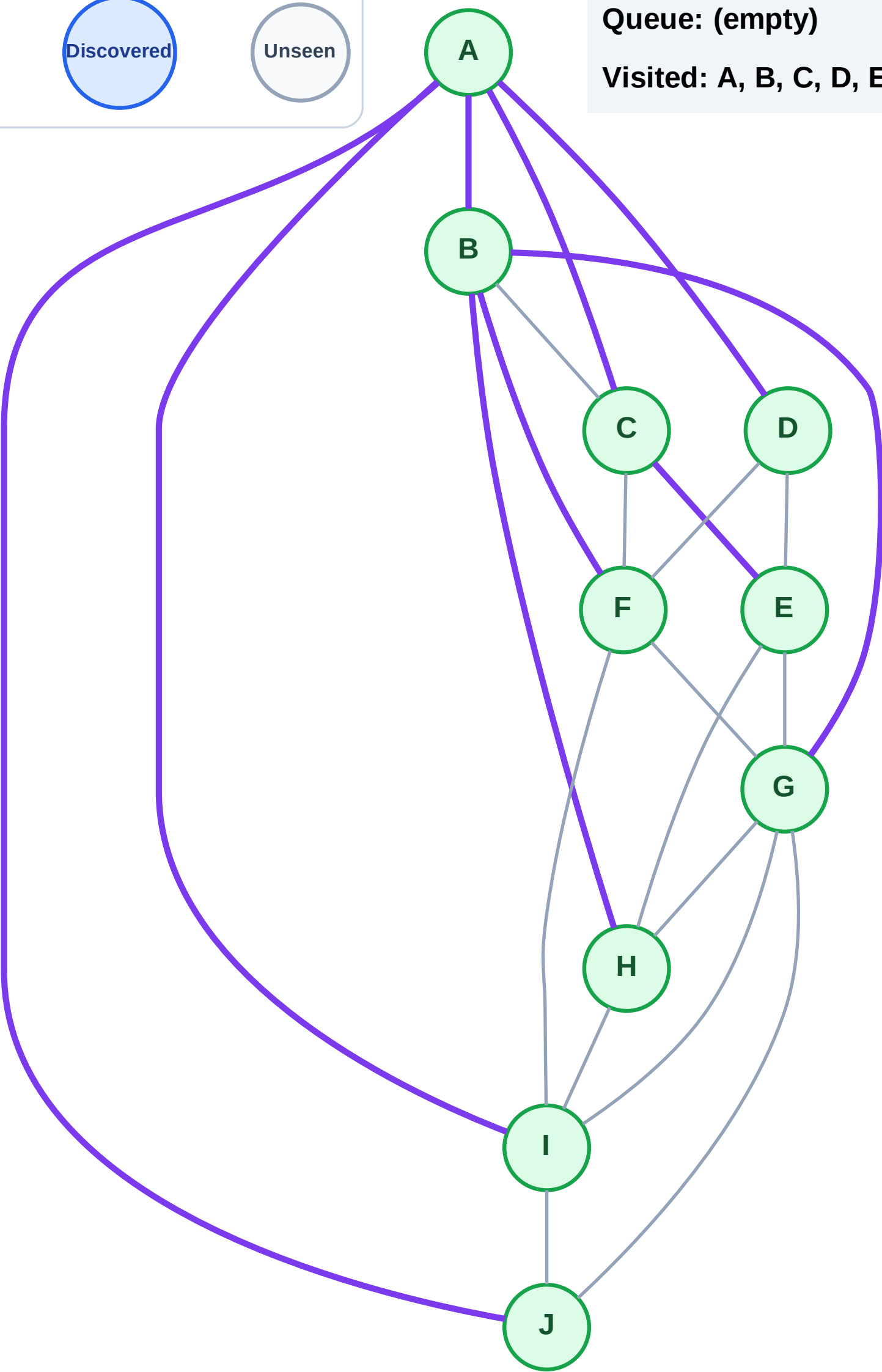
Step 21: No new neighbors from E

## Legend



Queue: (empty)

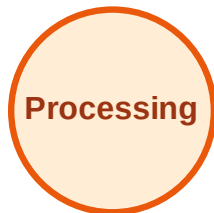
Visited: A, B, C, D, E, F, G, H, I, J



# DFS Traversal

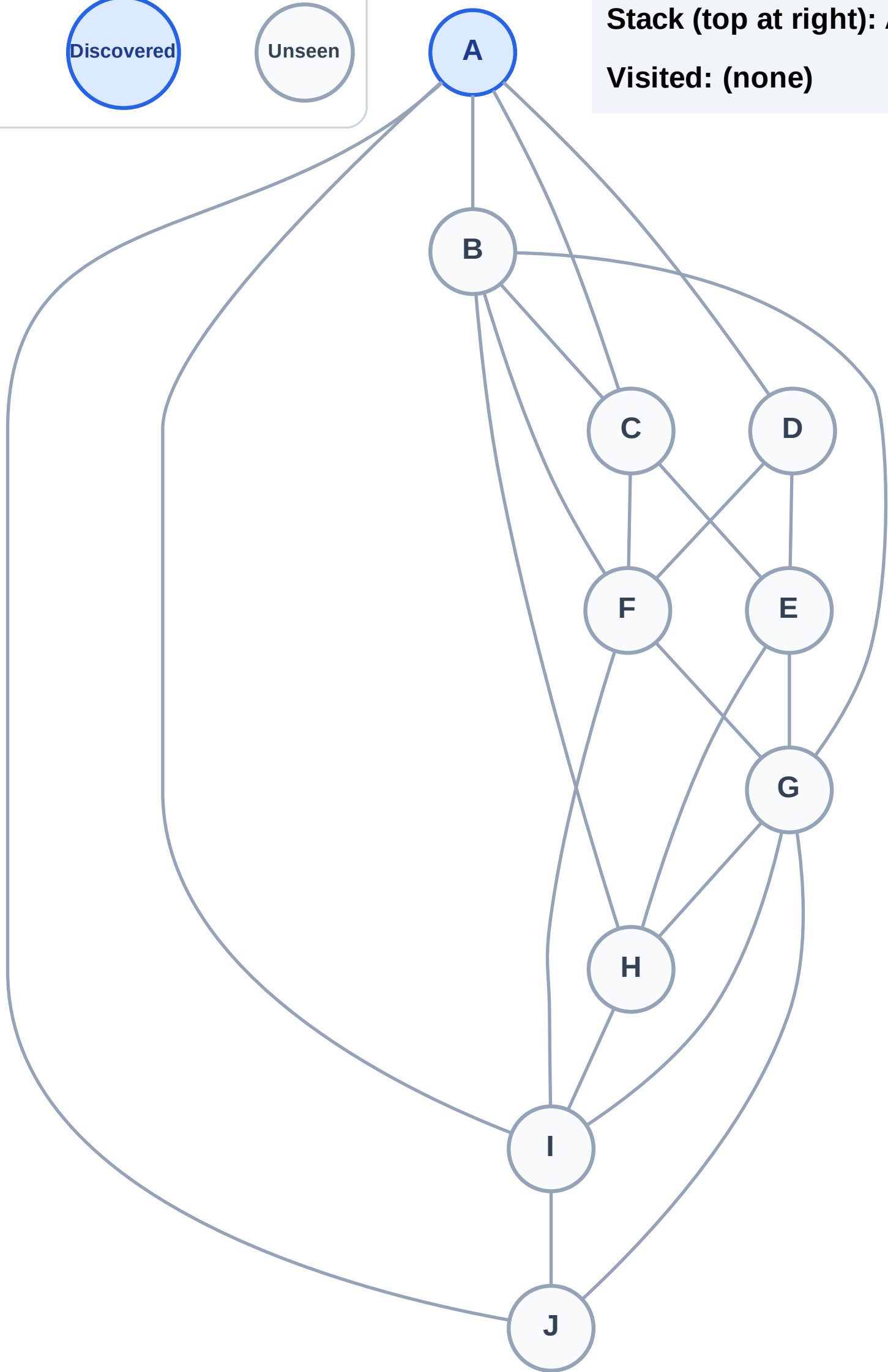
Step 1: Start at A

## Legend



Stack (top at right): A

Visited: (none)



# DFS Traversal

Step 2: Process node A

Legend

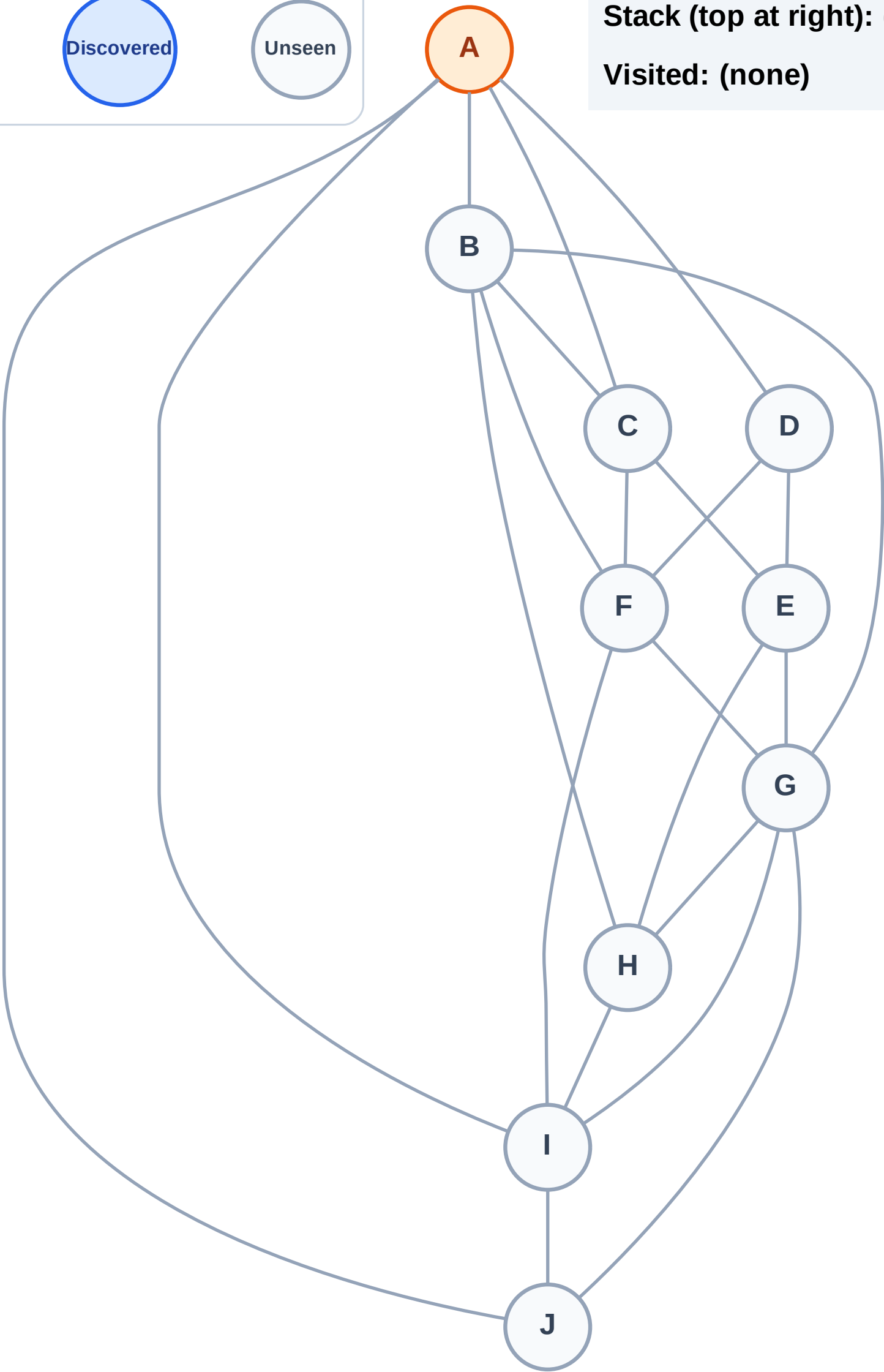
Complete

Processing

Discovered

Unseen

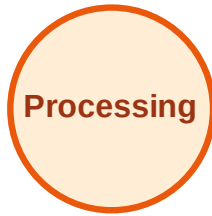
Stack (top at right): (empty)  
Visited: (none)



# DFS Traversal

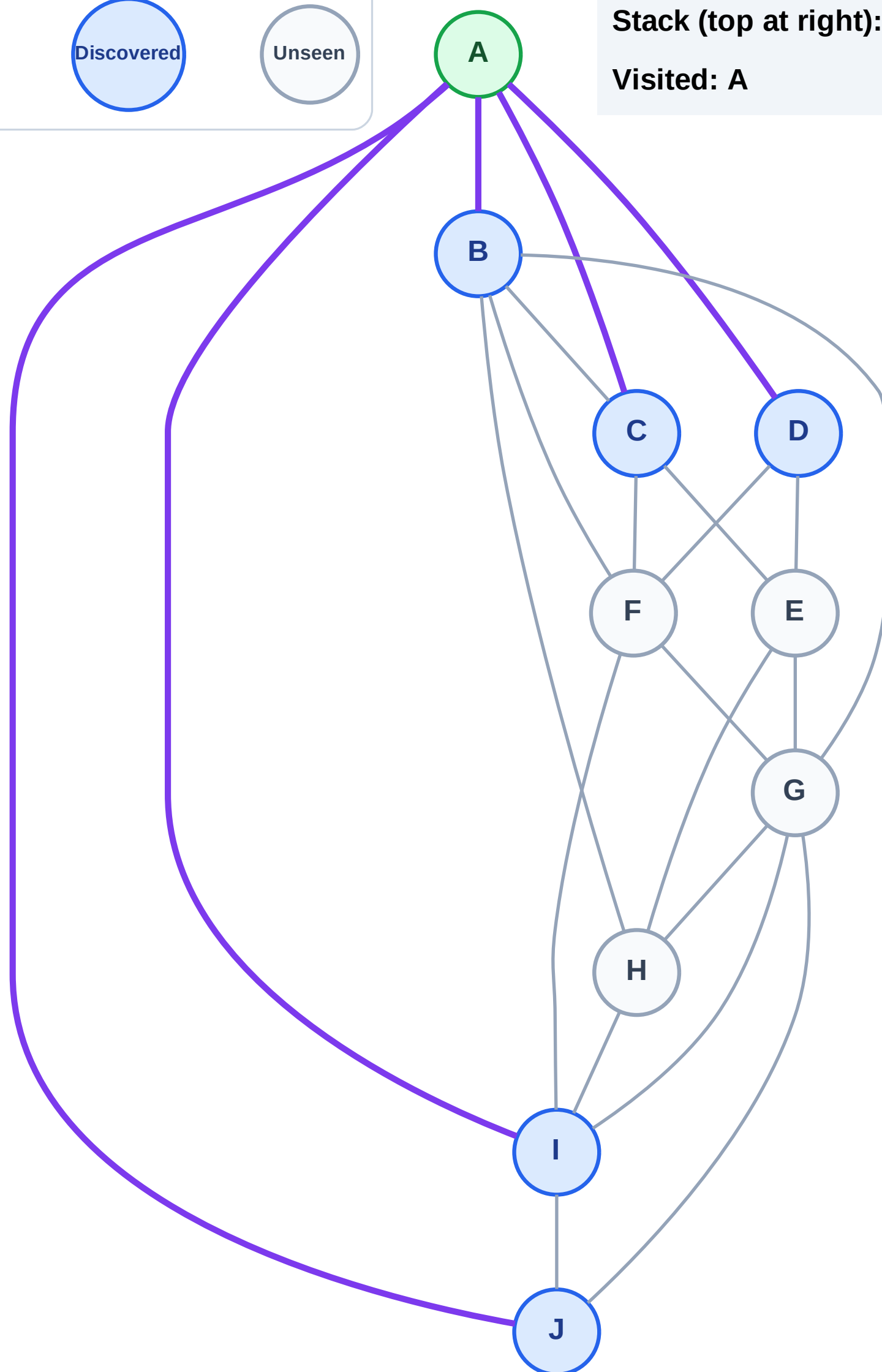
Step 3: Discover J, I, D, C, B from A

## Legend



Stack (top at right): J | I | D | C | B

Visited: A



# DFS Traversal

Step 4: Process node B

Legend

Complete

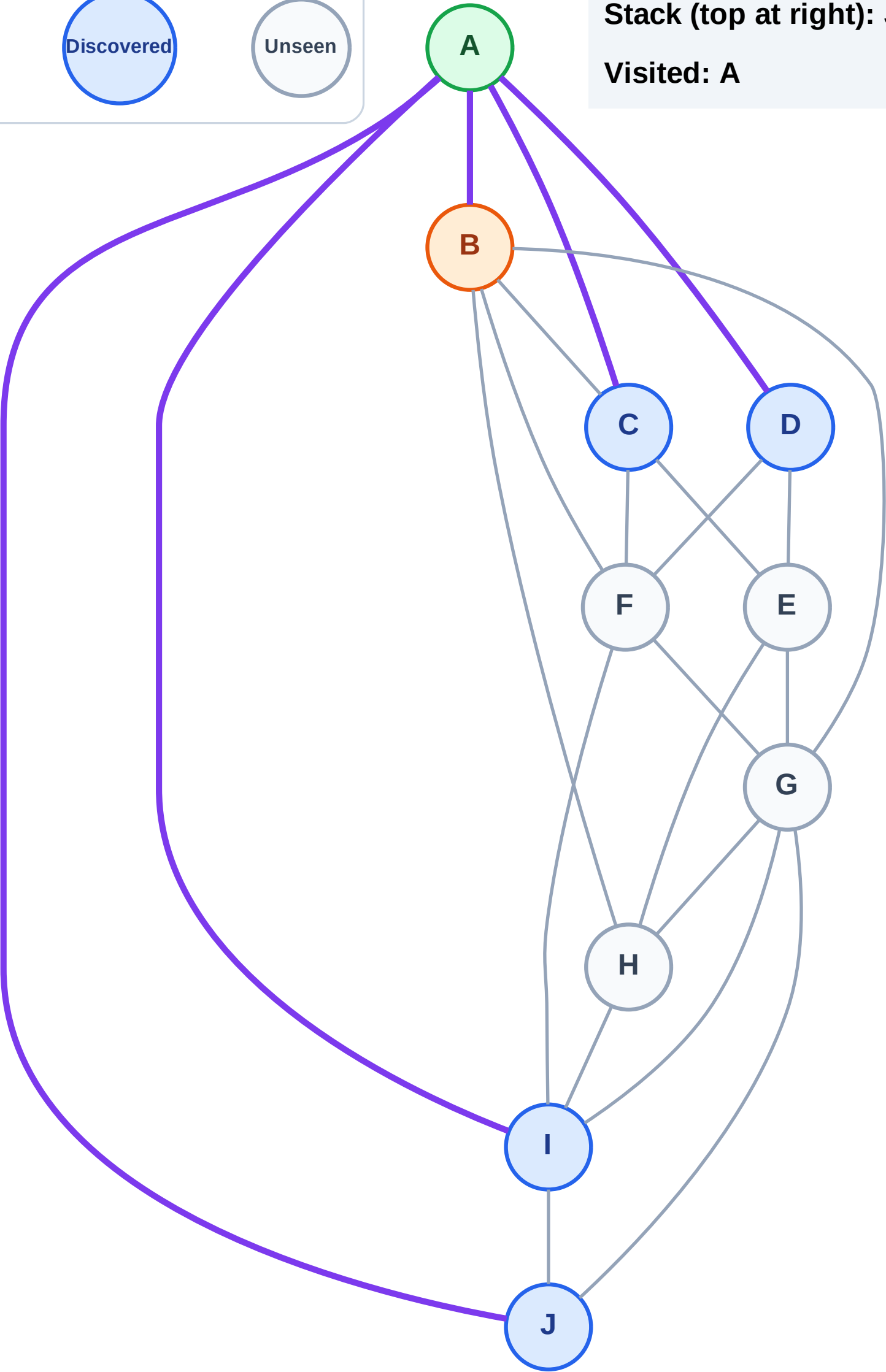
Processing

Discovered

Unseen

Stack (top at right): J | I | D | C

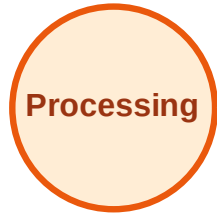
Visited: A



# DFS Traversal

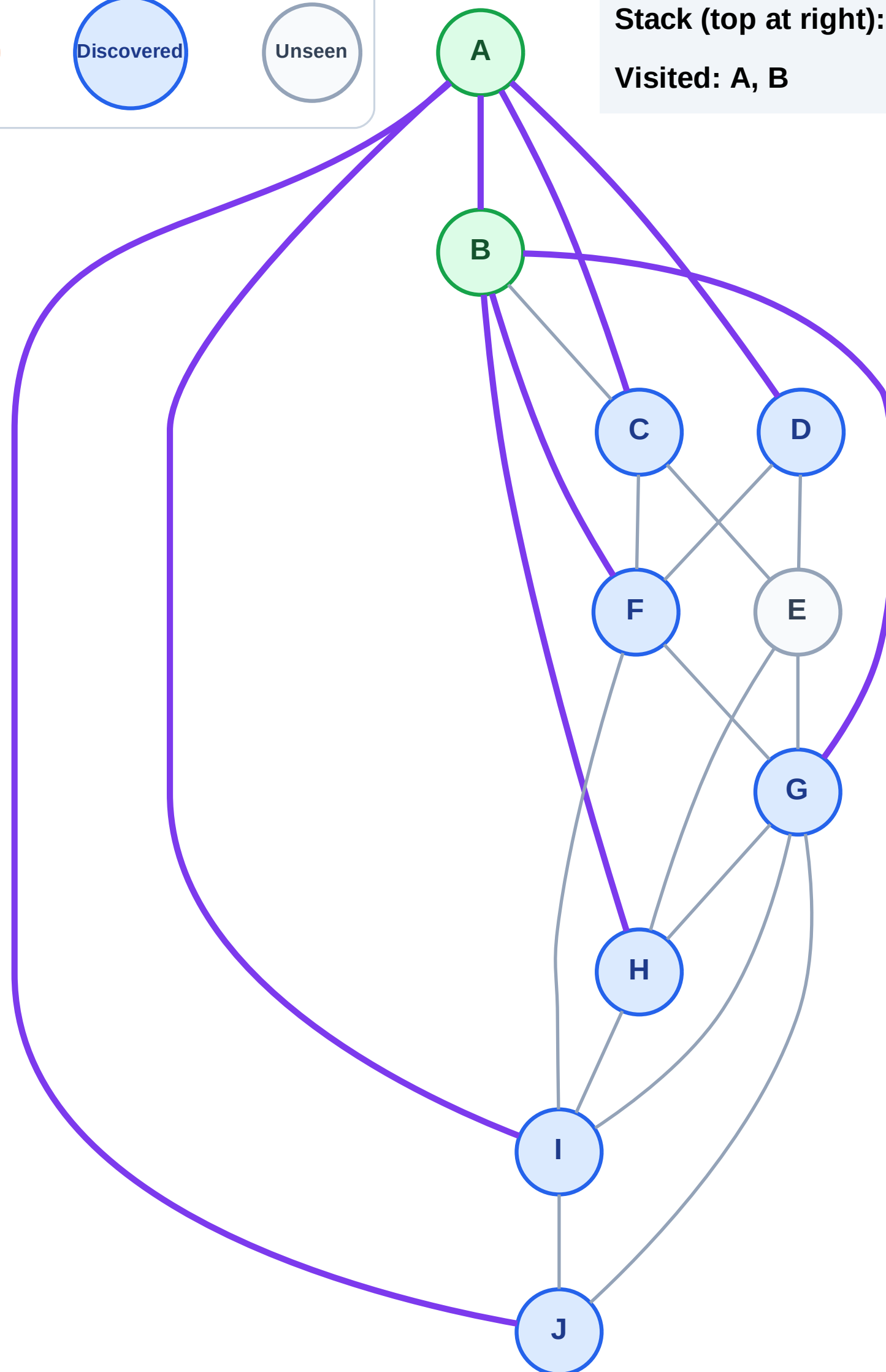
Step 5: Discover H, G, F from B

## Legend



Stack (top at right): J | I | D | C | H | G | F

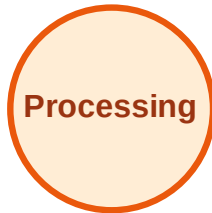
Visited: A, B



# DFS Traversal

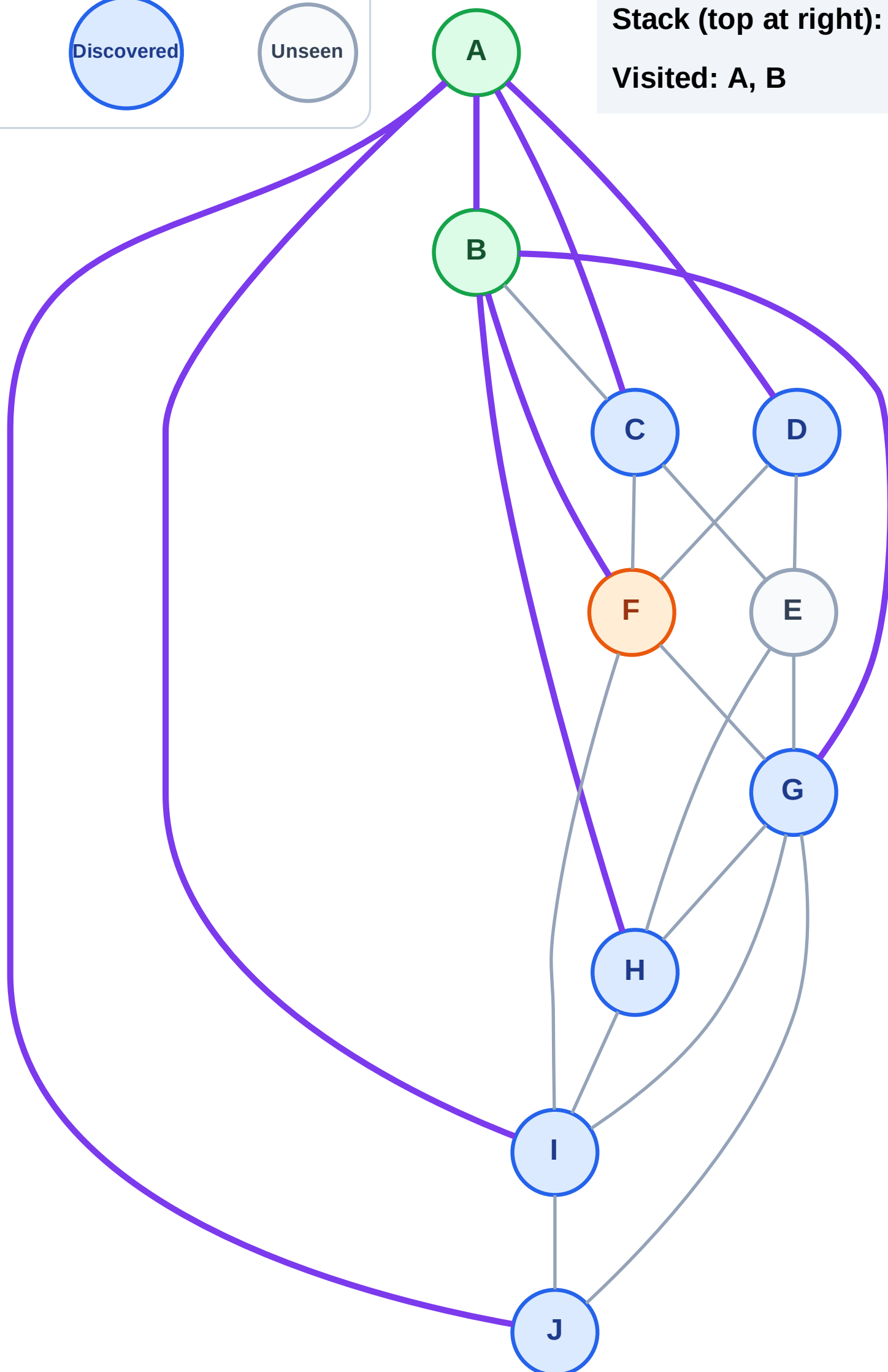
Step 6: Process node F

## Legend



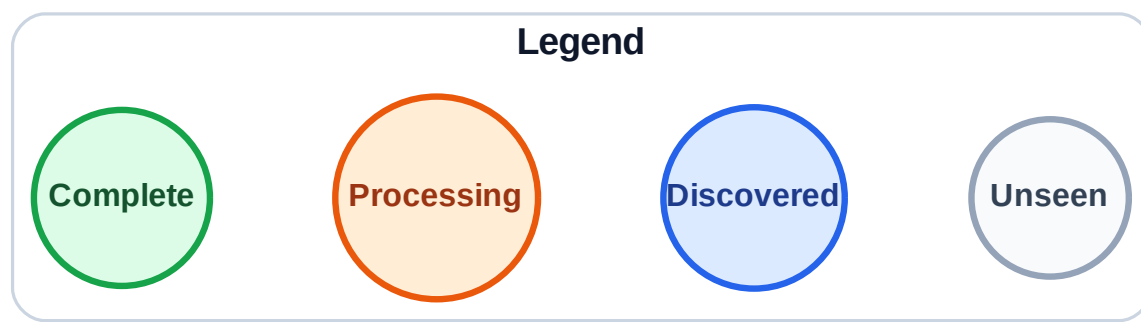
Stack (top at right): J | I | D | C | H | G

Visited: A, B



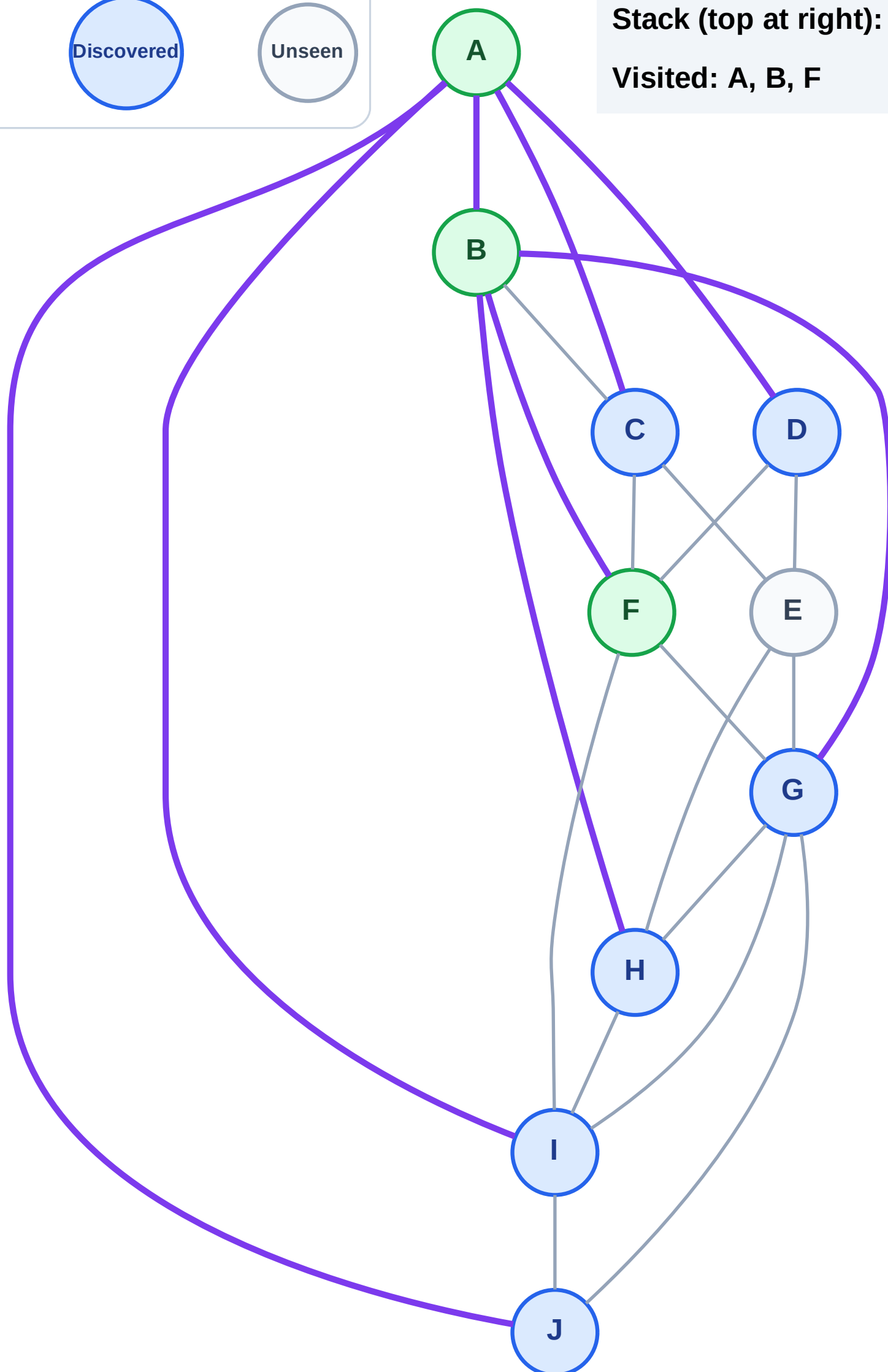
Step 7: No new neighbors from F

Step 7: No new neighbors from F



**Stack (top at right):** J | I | D | C | H | G

**Visited: A, B, F**



# DFS Traversal

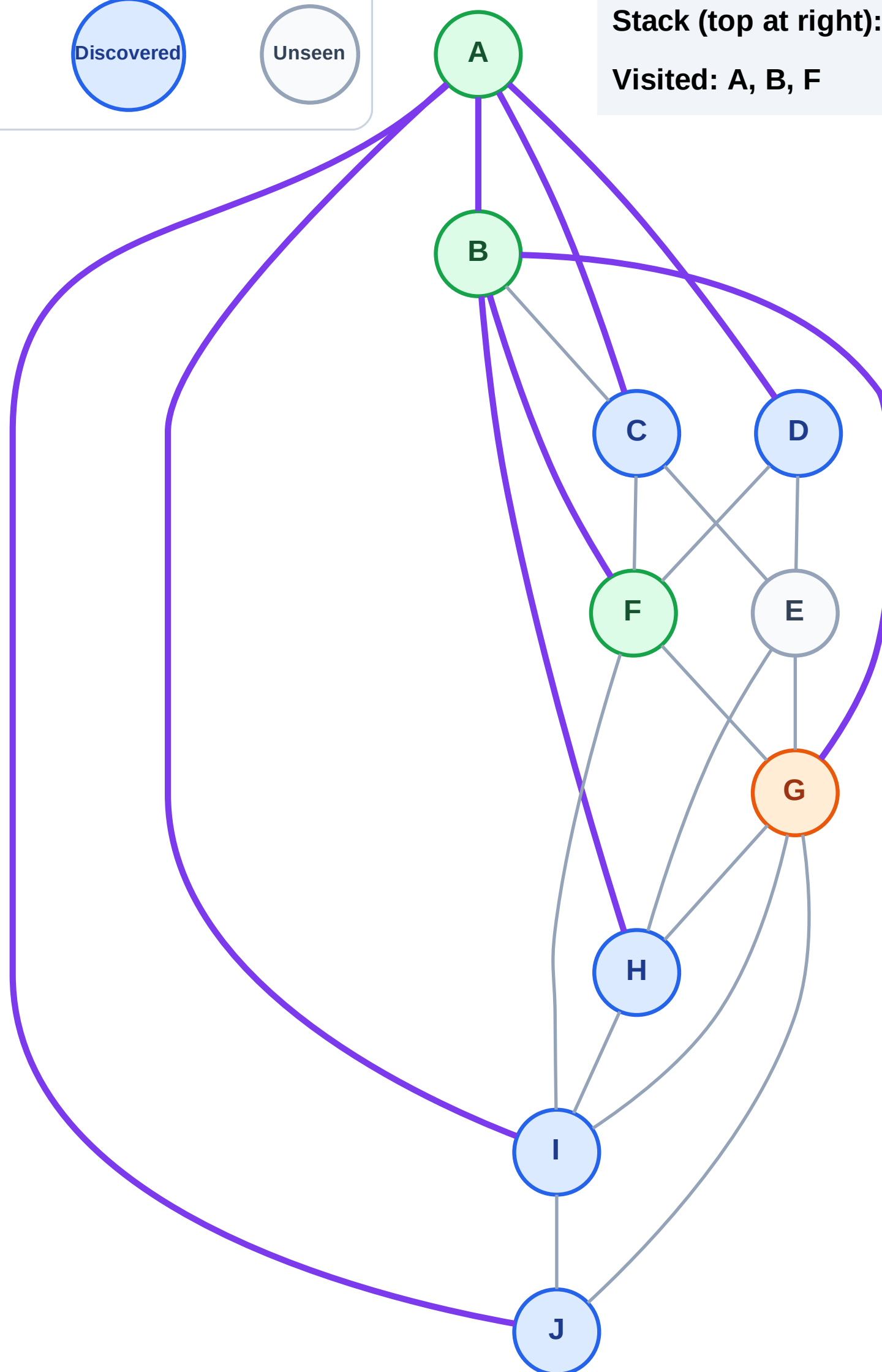
Step 8: Process node G

## Legend



Stack (top at right): J | I | D | C | H

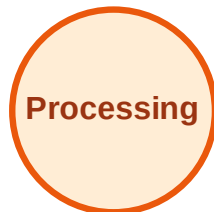
Visited: A, B, F



# DFS Traversal

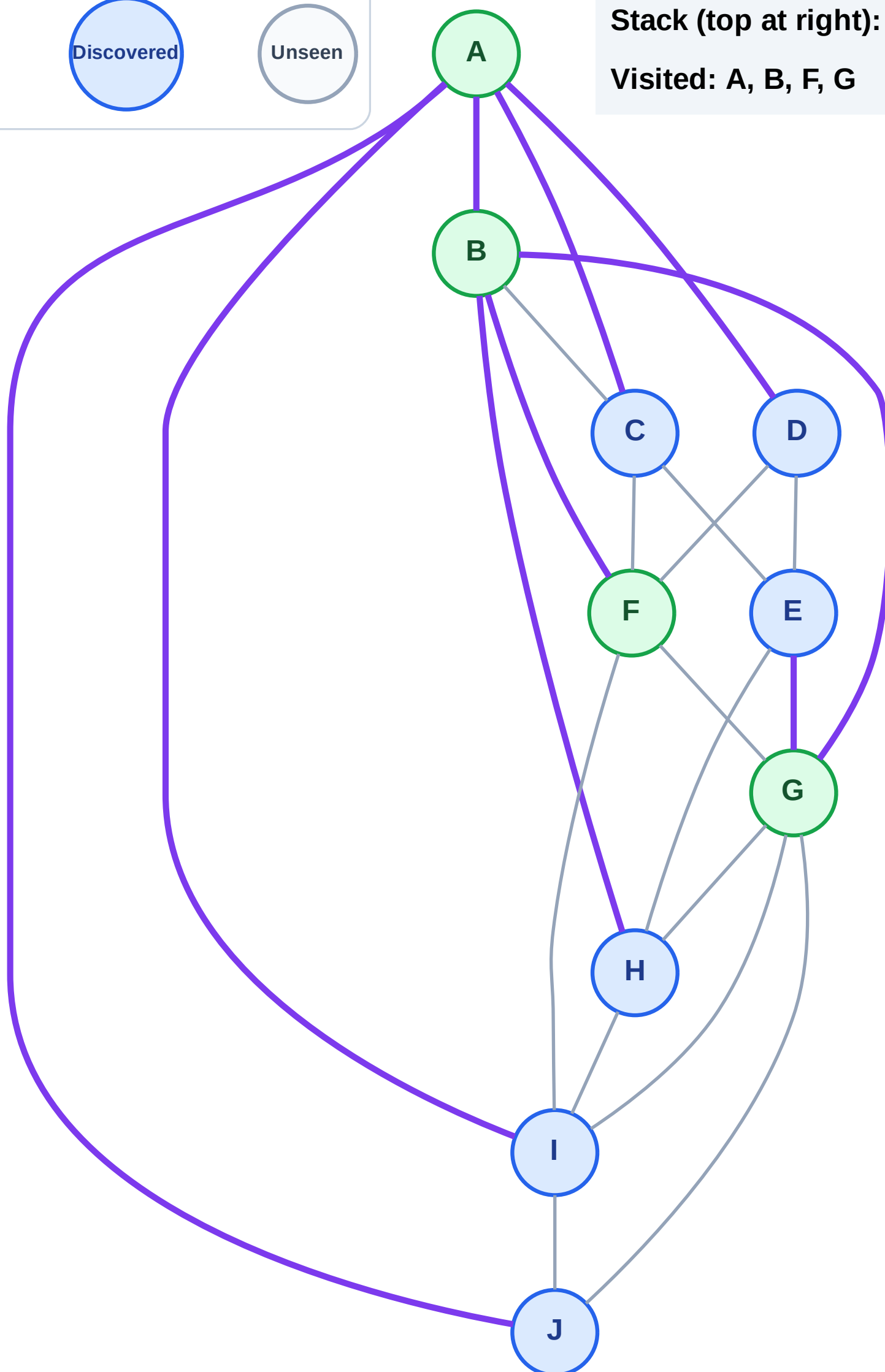
Step 9: Discover E from G

## Legend



Stack (top at right): J | I | D | C | H | E

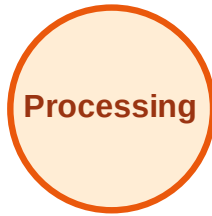
Visited: A, B, F, G



# DFS Traversal

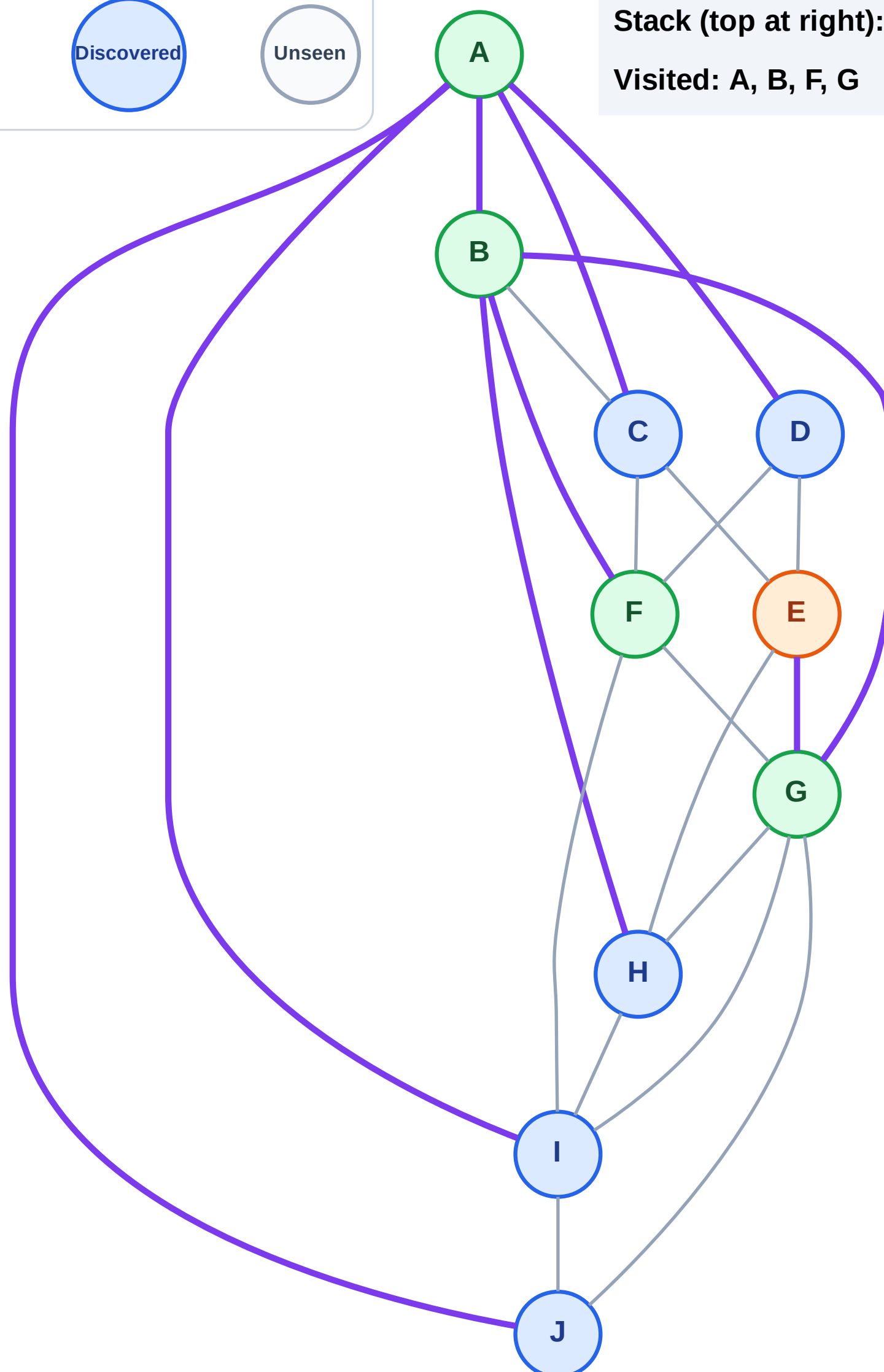
Step 10: Process node E

## Legend



Stack (top at right): J | I | D | C | H

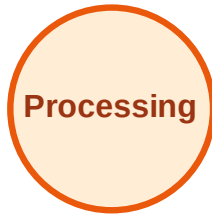
Visited: A, B, F, G



# DFS Traversal

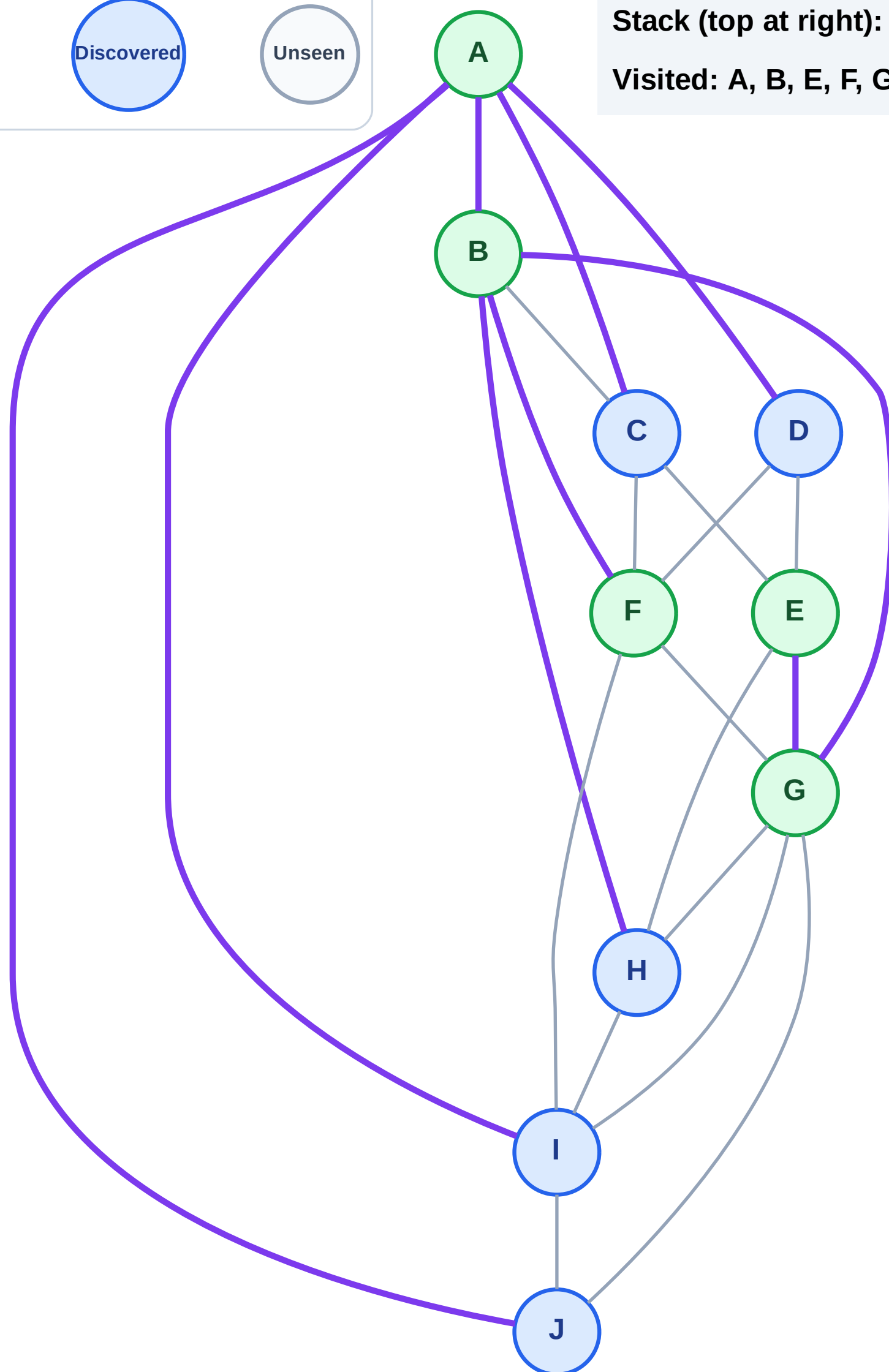
Step 11: No new neighbors from E

## Legend



Stack (top at right): J | I | D | C | H

Visited: A, B, E, F, G



# DFS Traversal

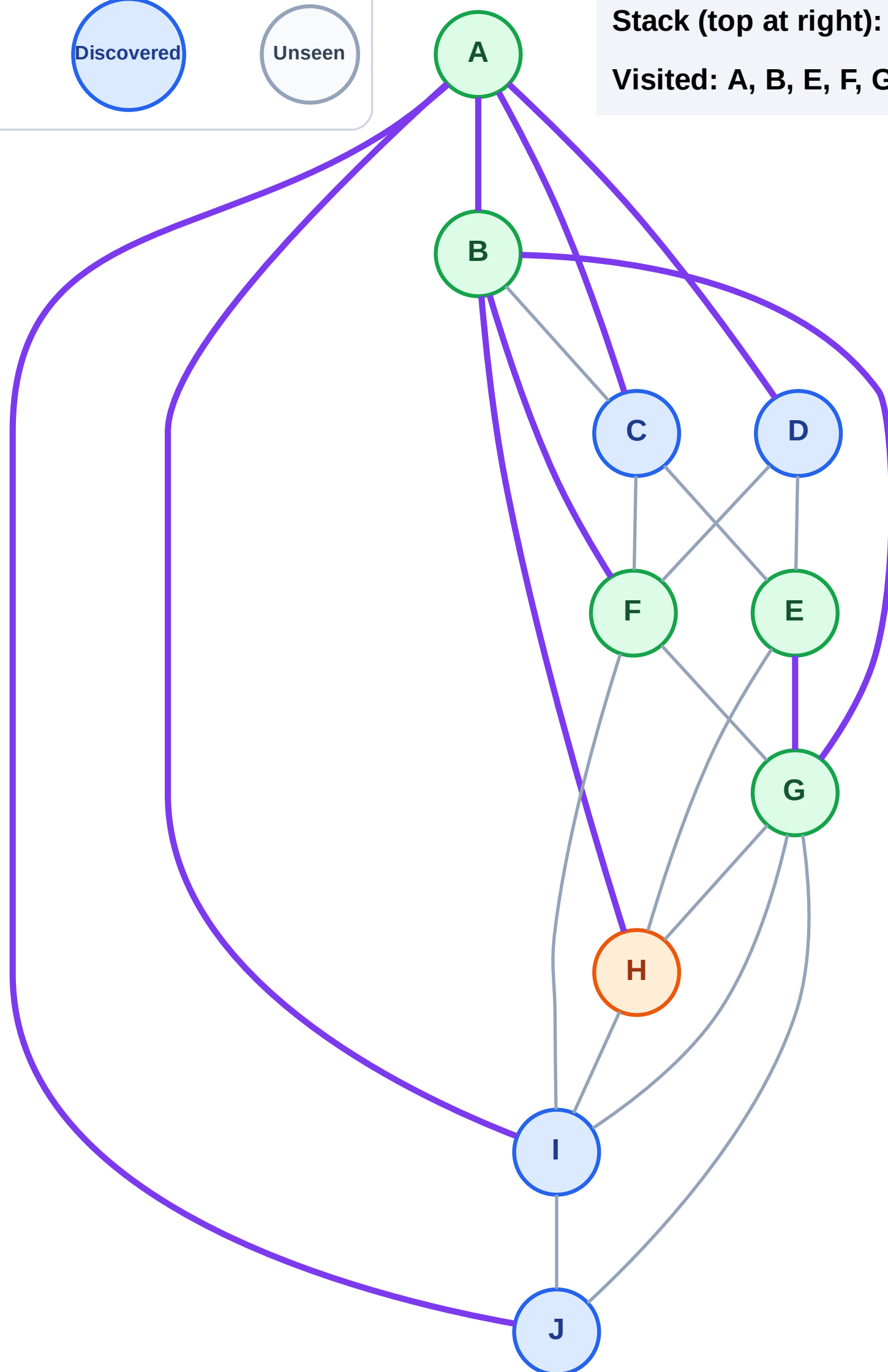
Step 12: Process node H

## Legend



Stack (top at right): J | I | D | C

Visited: A, B, E, F, G



# DFS Traversal

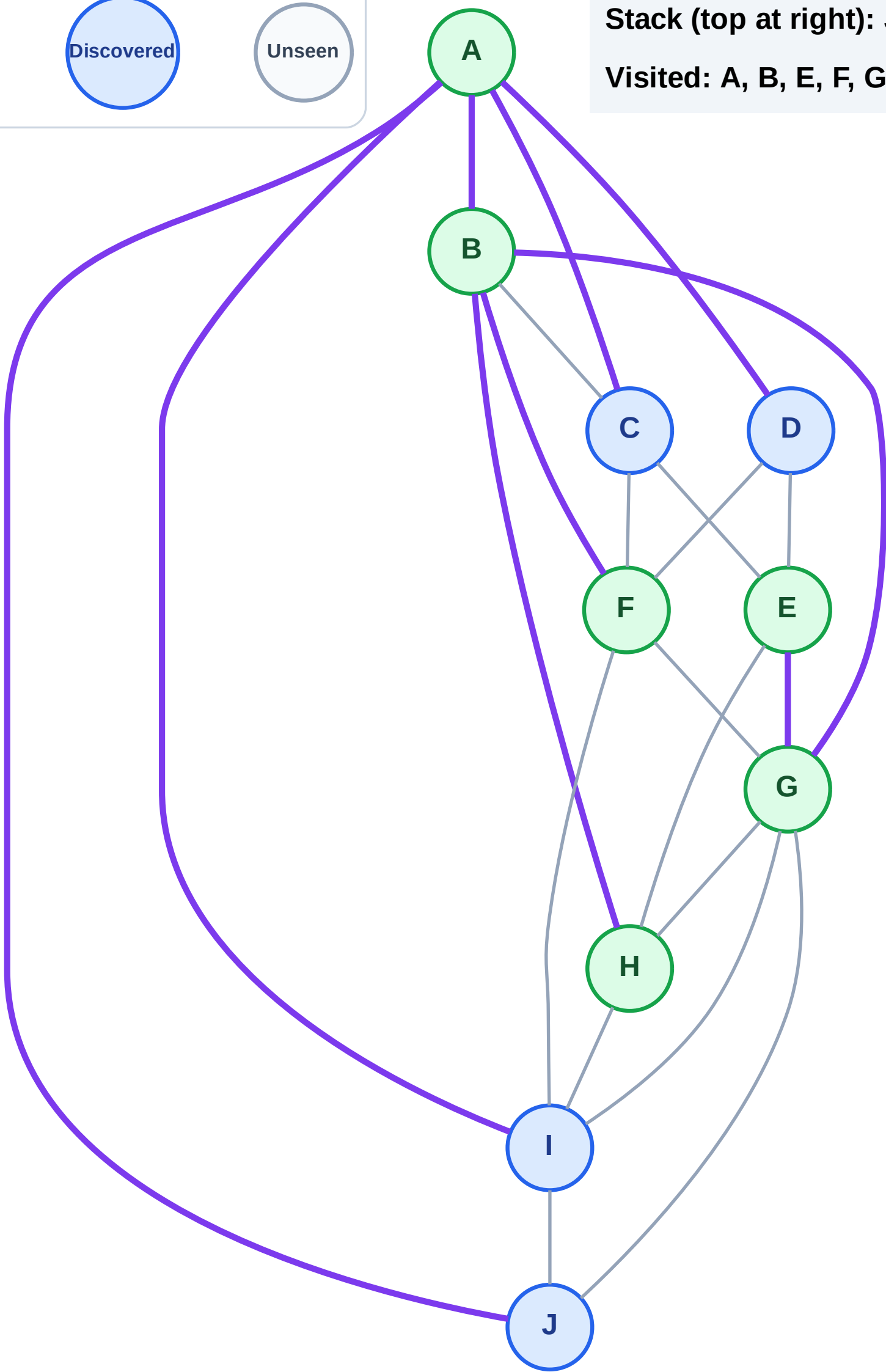
Step 13: No new neighbors from H

## Legend



Stack (top at right): J | I | D | C

Visited: A, B, E, F, G, H



# DFS Traversal

Step 14: Process node C

## Legend

Complete

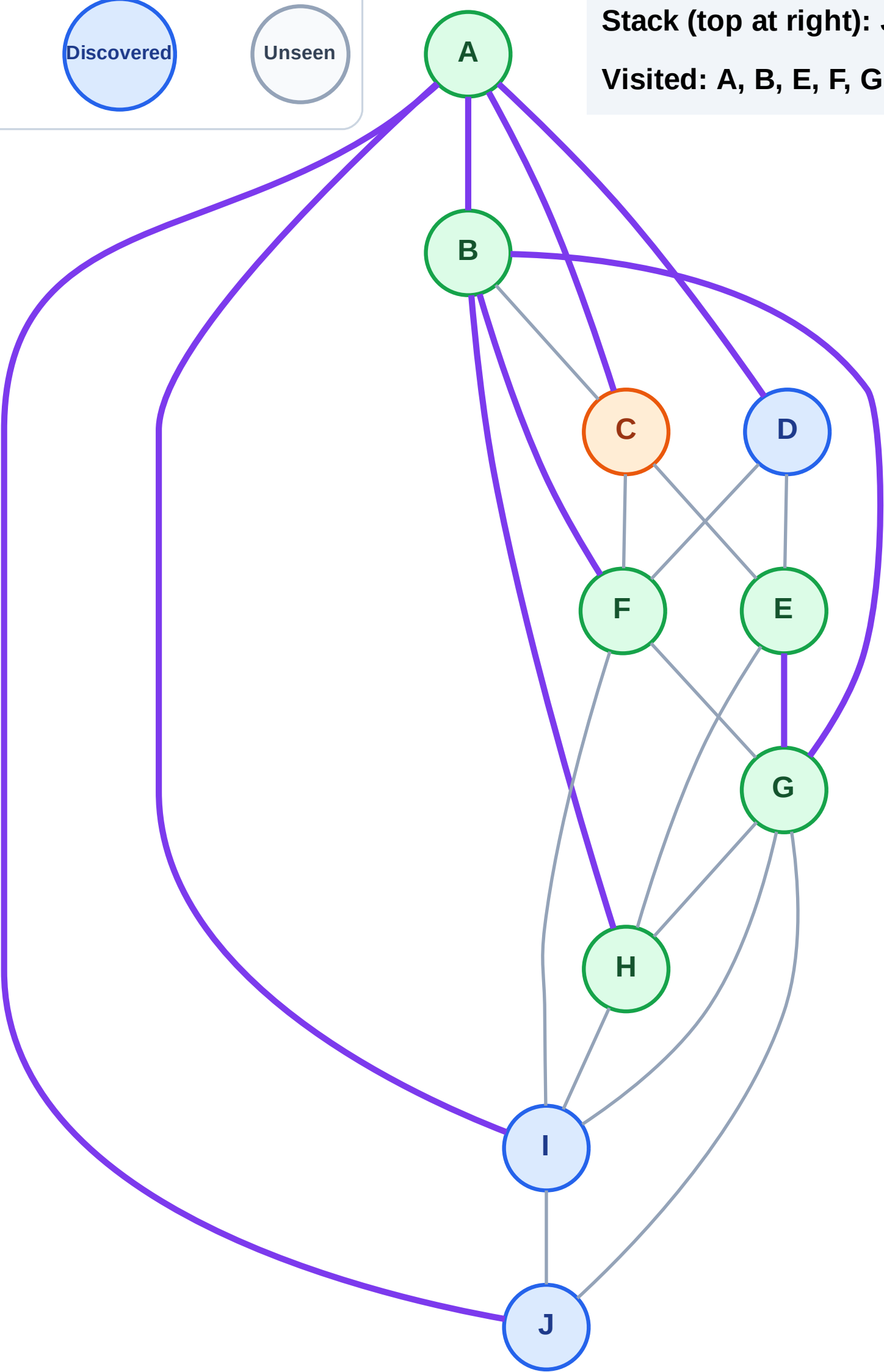
Processing

Discovered

Unseen

Stack (top at right): J | I | D

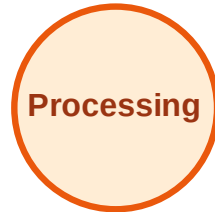
Visited: A, B, E, F, G, H



# DFS Traversal

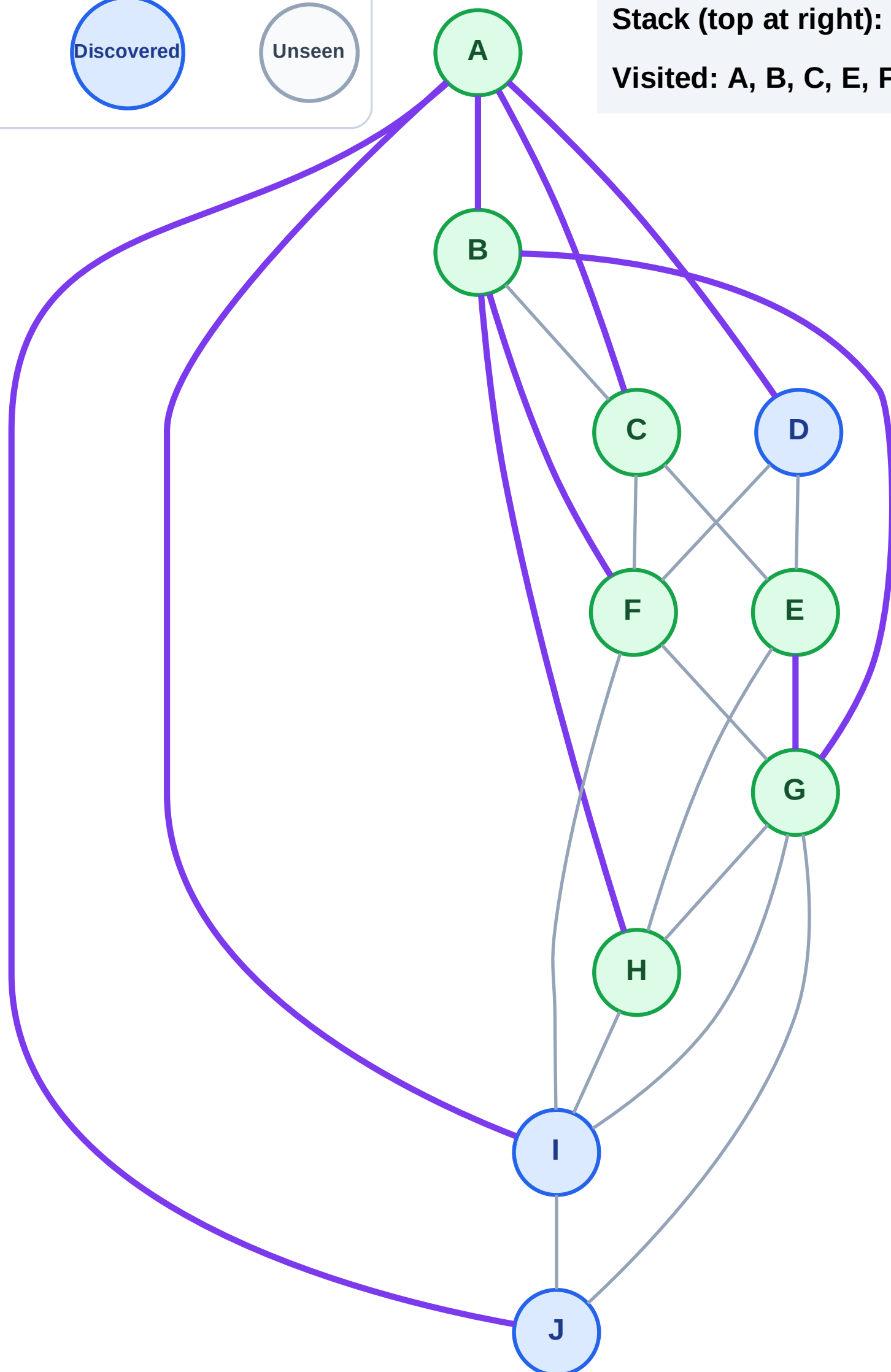
Step 15: No new neighbors from C

## Legend



Stack (top at right): J | I | D

Visited: A, B, C, E, F, G, H



# DFS Traversal

Step 16: Process node D

## Legend

Complete

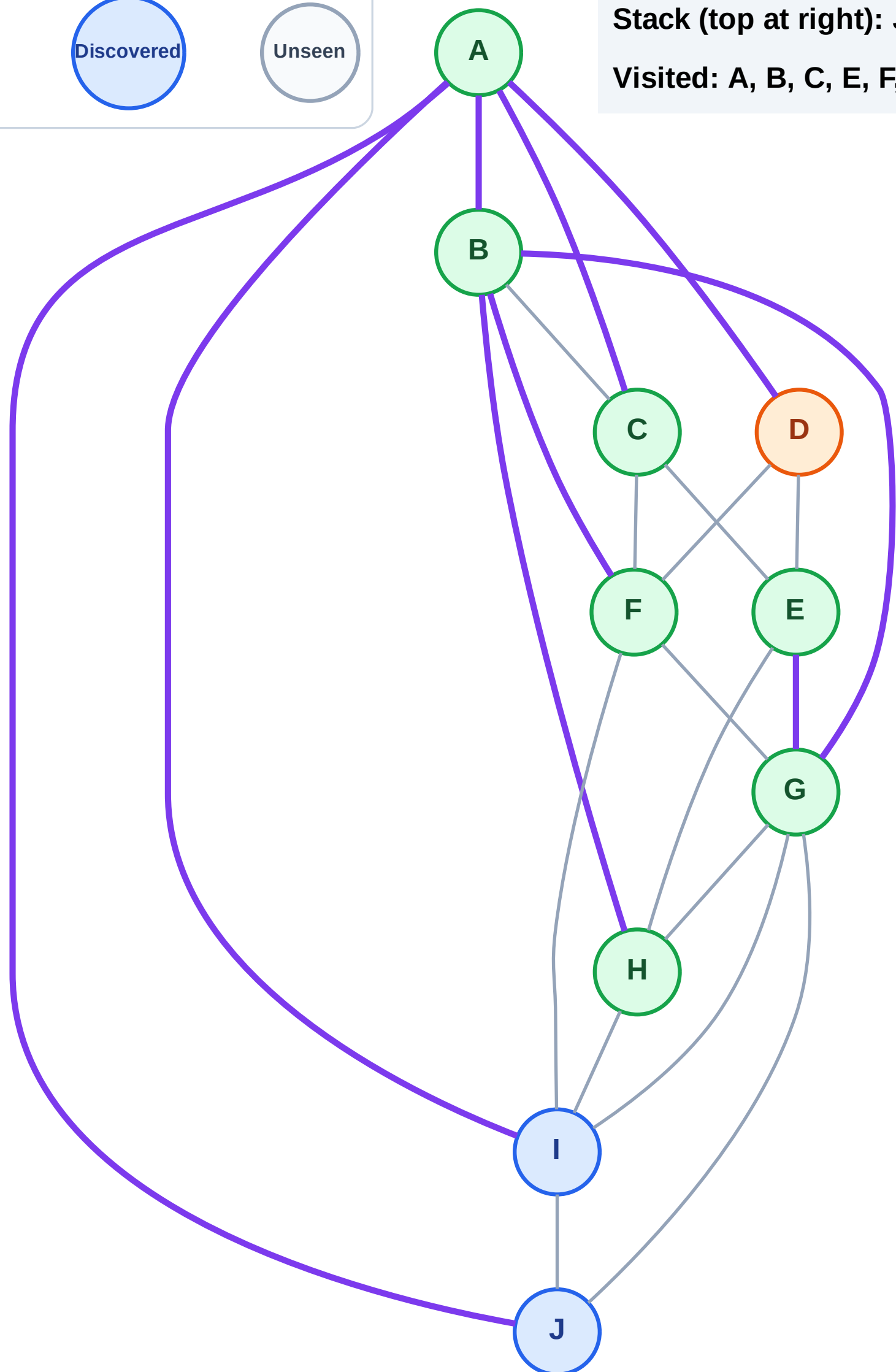
Processing

Discovered

Unseen

Stack (top at right): J | I

Visited: A, B, C, E, F, G, H



# DFS Traversal

Step 17: No new neighbors from D

## Legend

Complete

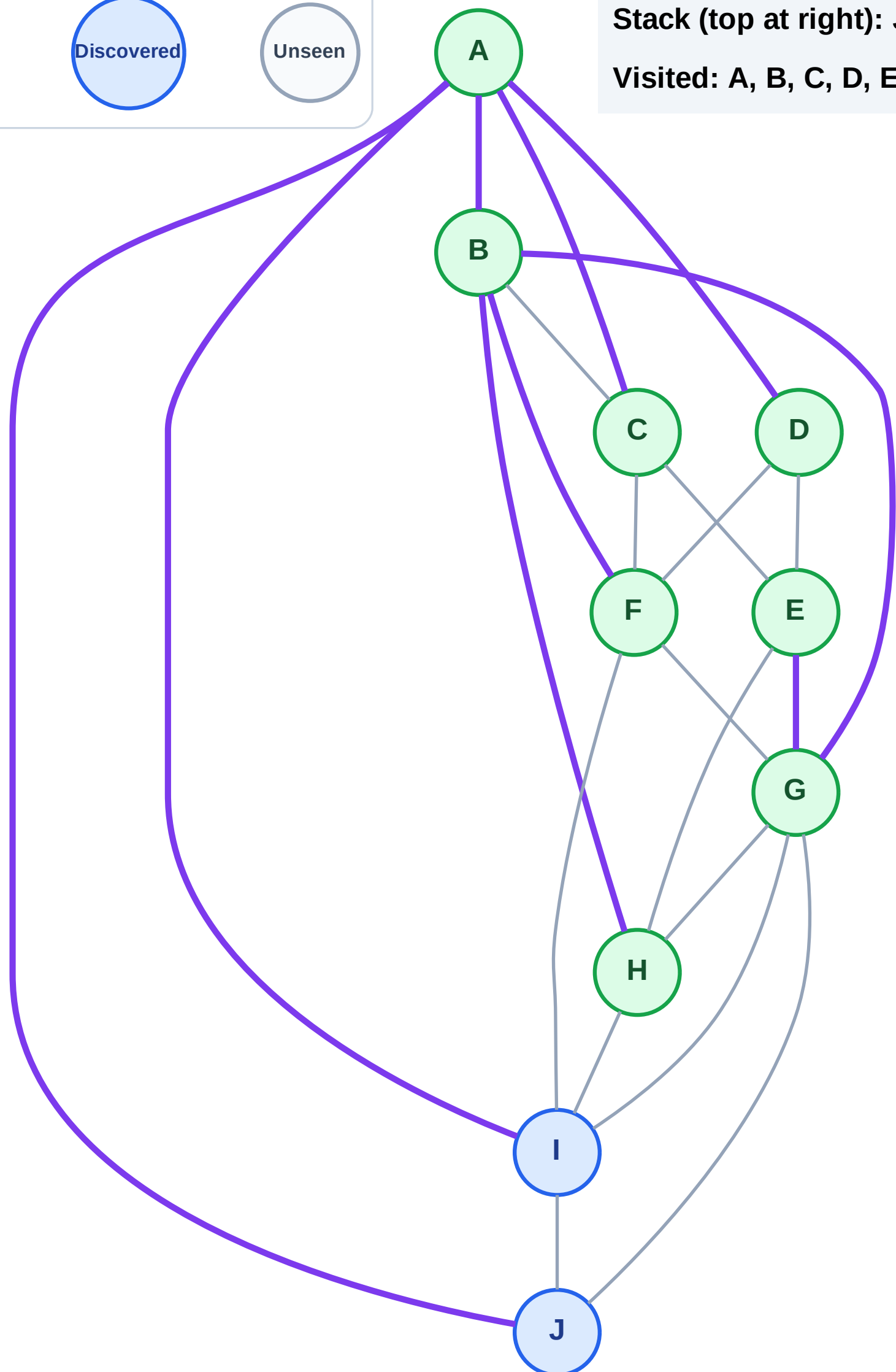
Processing

Discovered

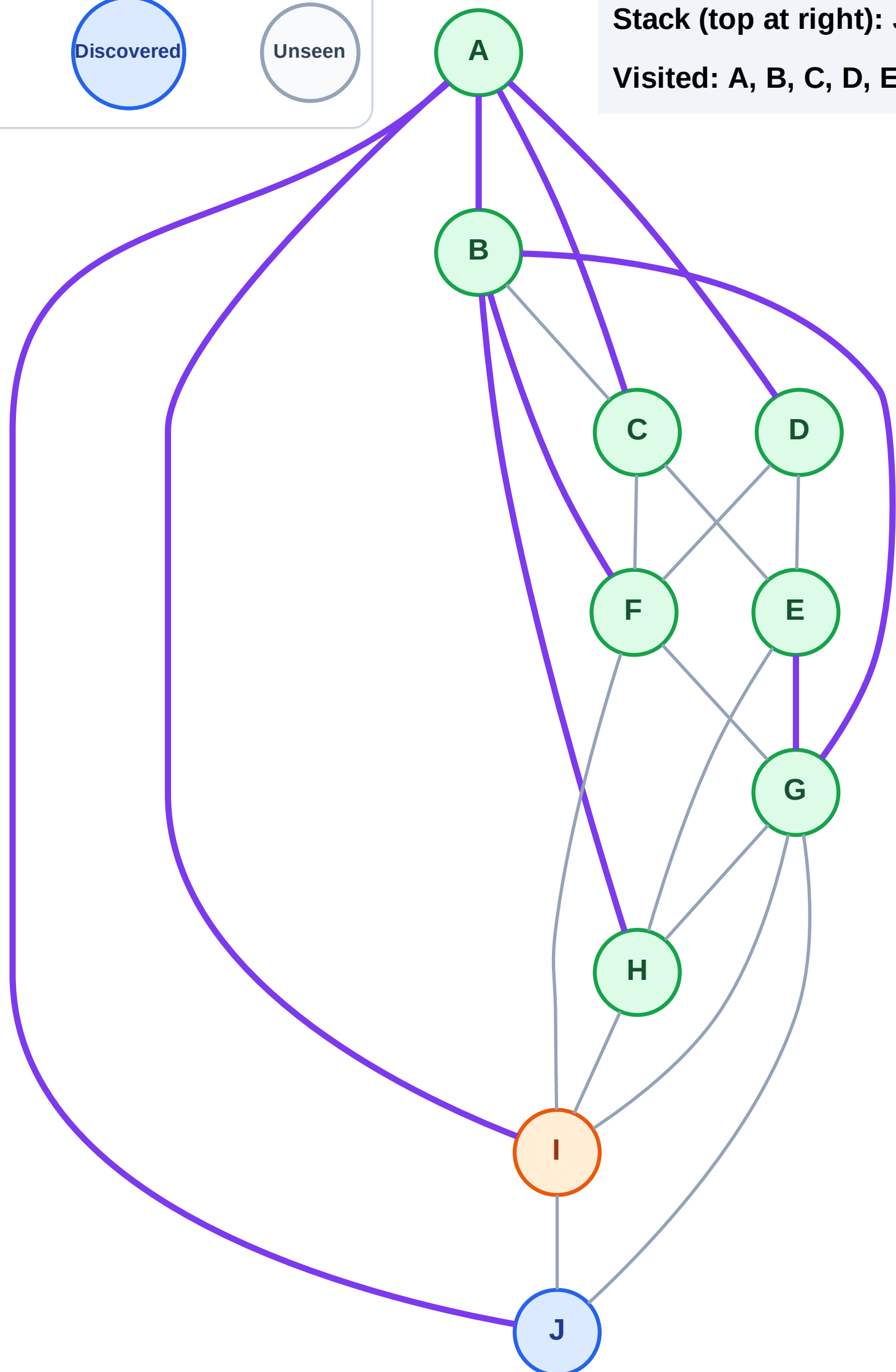
Unseen

Stack (top at right): J | I

Visited: A, B, C, D, E, F, G, H



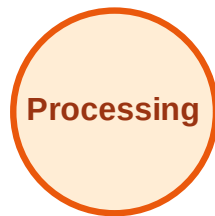
### Step 18: Process node I



# DFS Traversal

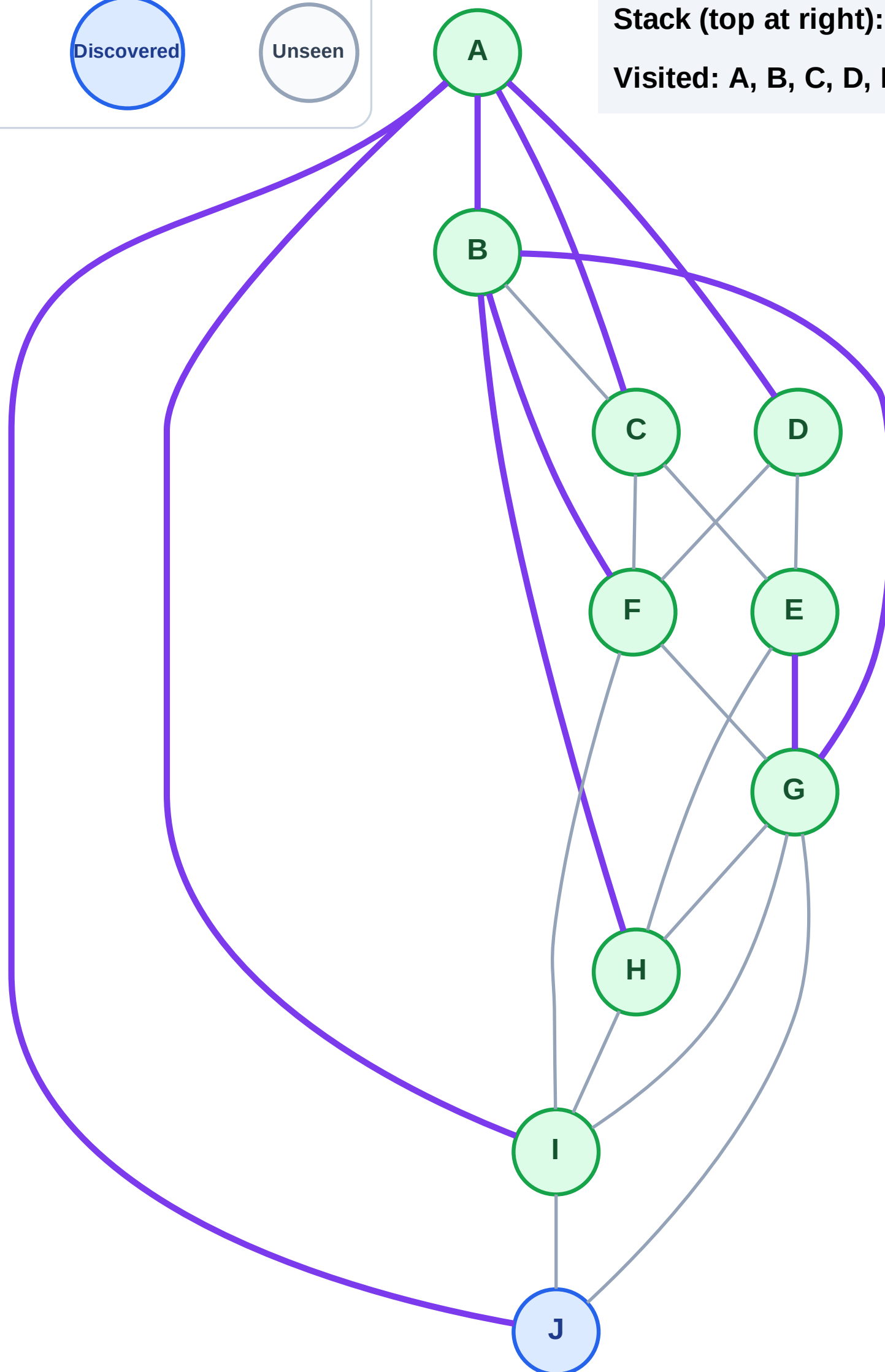
Step 19: No new neighbors from I

## Legend



Stack (top at right): J

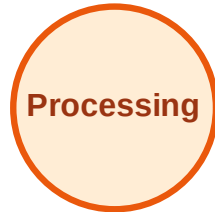
Visited: A, B, C, D, E, F, G, H, I



# DFS Traversal

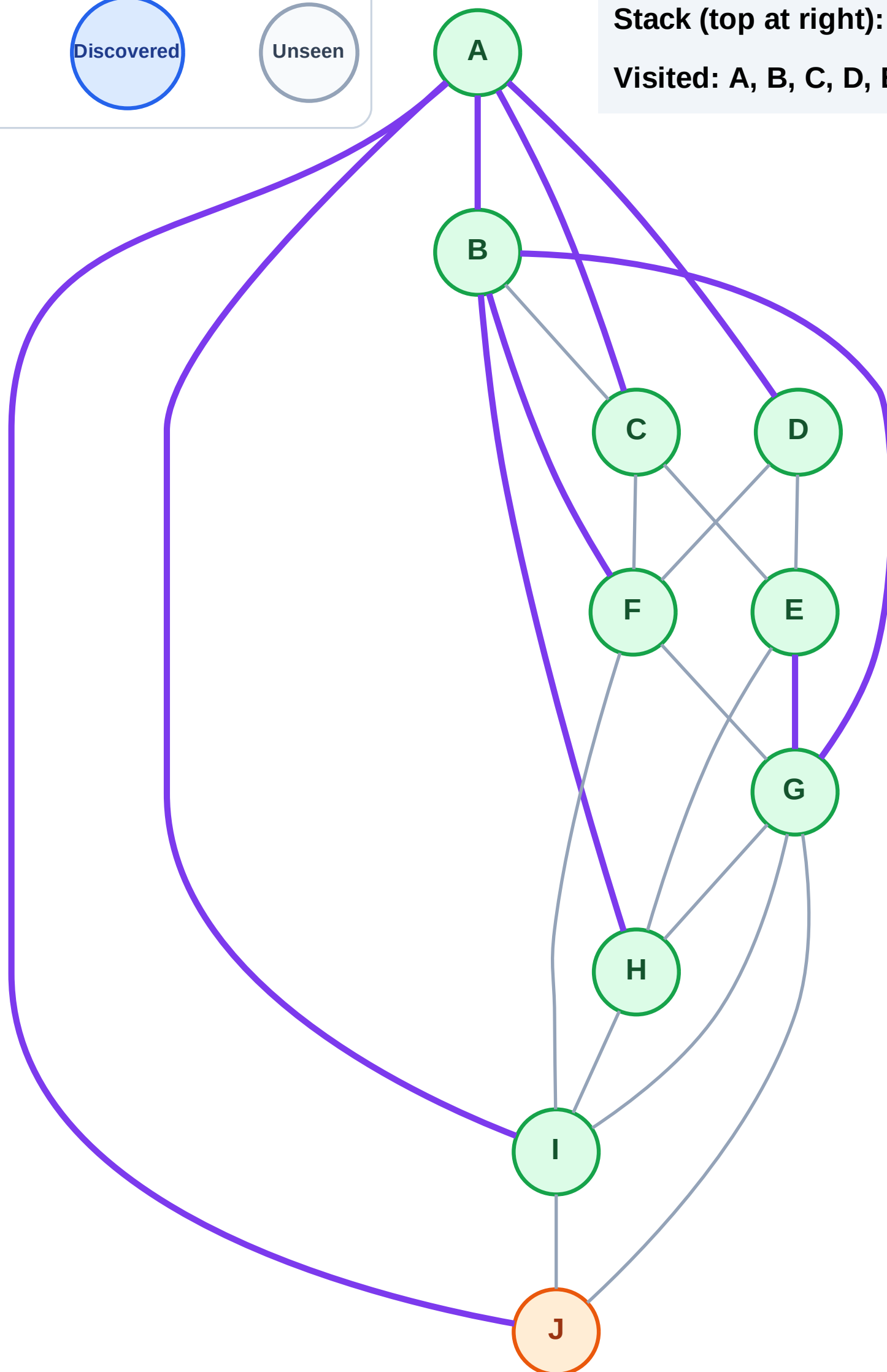
Step 20: Process node J

## Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I



# DFS Traversal

Step 21: No new neighbors from J

## Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

